

Big Data Course Overview

Juliana Freire

Computer Science & Engineering
Visualization and Data Analysis (ViDA)
Center for Data Science (CDS)



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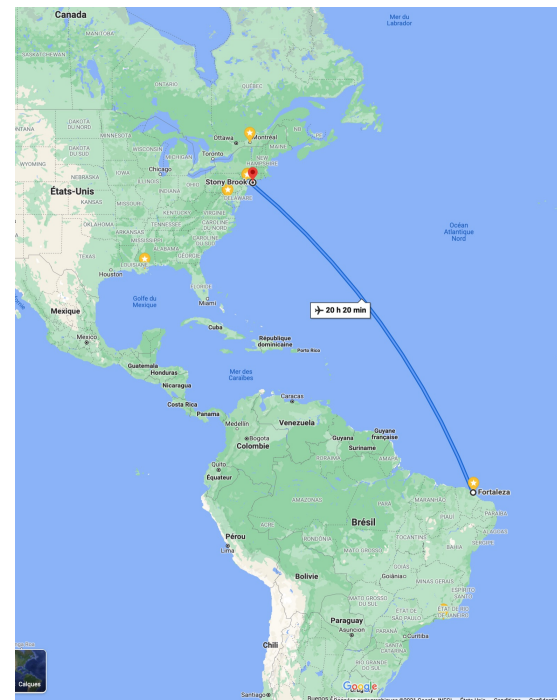


Course Staff

- Instructor: Juliana Freire, Professor of Computer Science and Data Science
- Course Assistant: TBD

My Journey

- B.S. Computer Science UFC (Brazil) → SUNY at Stony Brook



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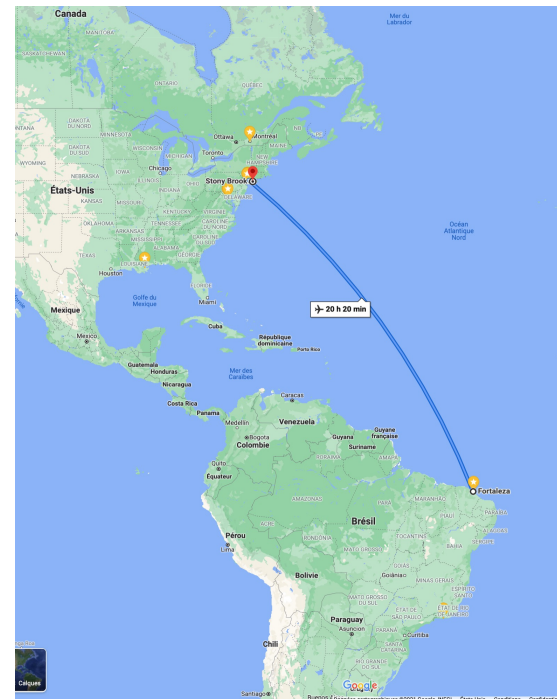
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My Journey

- B.S. Computer Science UFC (Brazil) → SUNY at Stony Brook
- MS and PhD Computer Science, SUNY at Stony Brook



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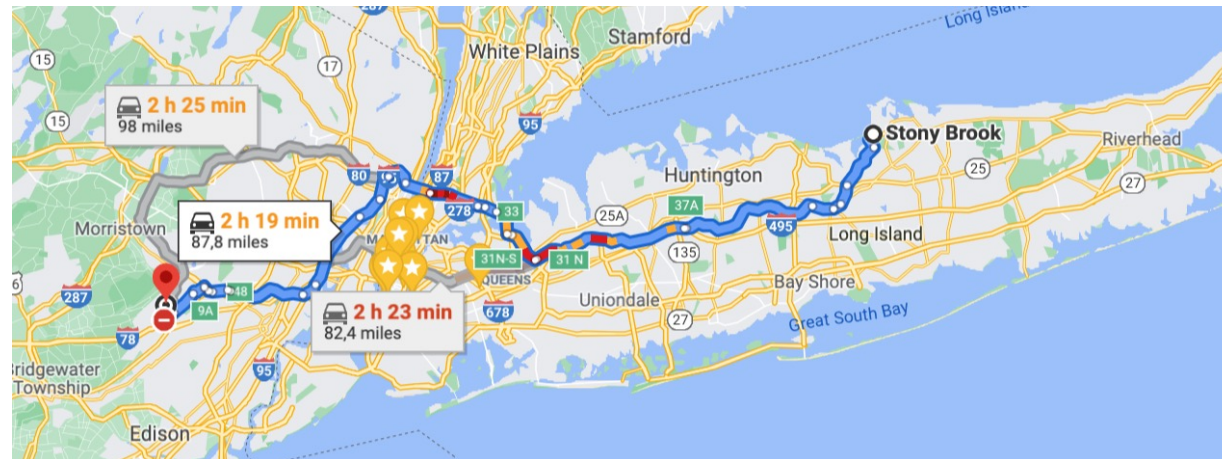
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My Journey

- UFC (Brazil) →
SUNY at Stony Brook →
Bell Labs



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My Journey

Some of Bell Labs inventions:

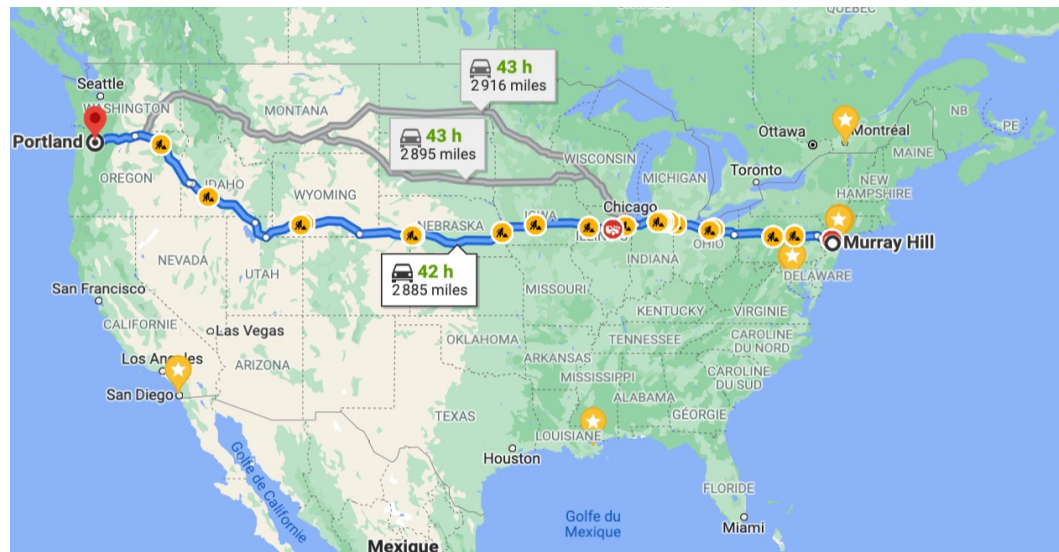
- Transistor: Invented in 1947
- Laser: Invented in 1958
- Solar cell: Invented in 1954
- Information theory: Invented in 1948
- Cellular concept: Invented in 1947
- Telstar 1 satellite: Launched in 1962
- Charged-coupled device (CCD): Invented in 1969
- UNIX operating system: Invented in 1969
- Programming languages: B, C, C++, S, SNOBOL, and AWK



“work hard and be famous”
Narain Gehani

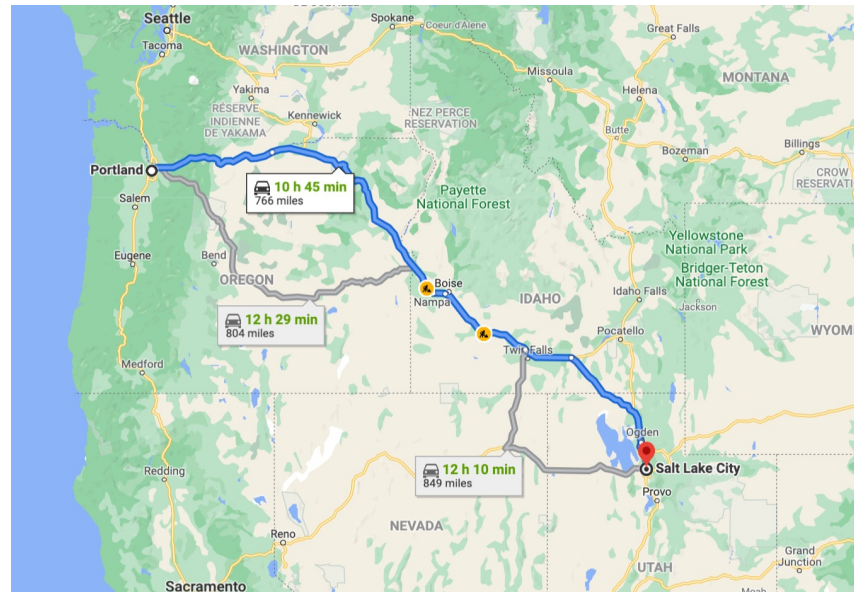
My Journey

- UFC (Brazil) →
SUNY at Stony Brook →
Bell Labs →
OGI/OHSU



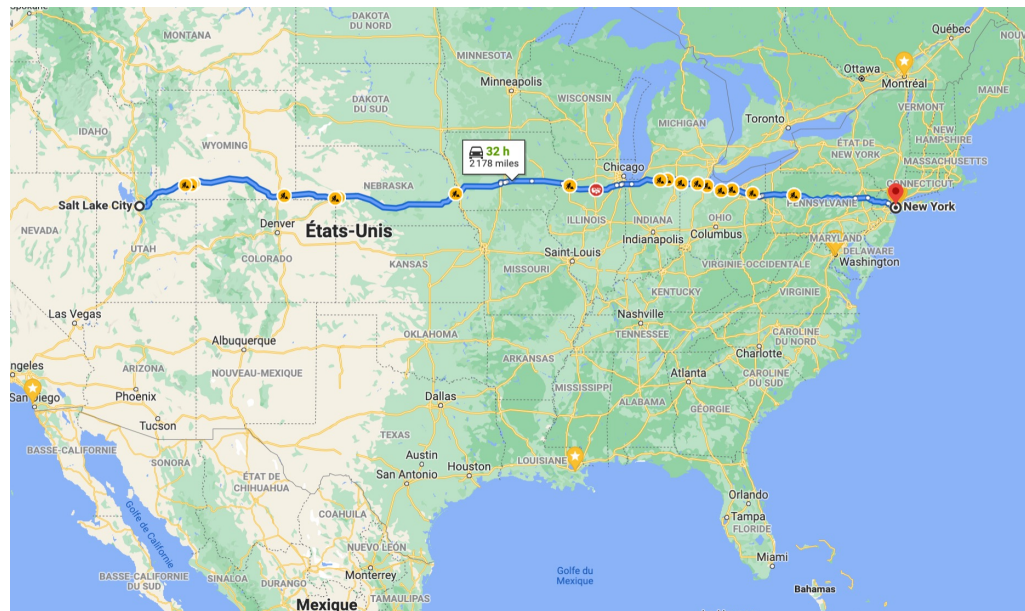
My Journey

- UFC (Brazil) →
SUNY at Stony Brook →
Bell Labs →
OGI/OHSU →
U of Utah



My Journey

- UFC (Brazil) →
SUNY at Stony Brook →
Bell Labs →
OGI/OHSU →
U of Utah →
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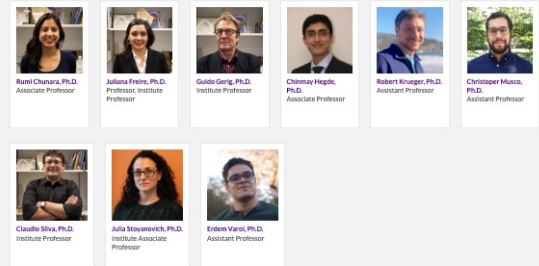
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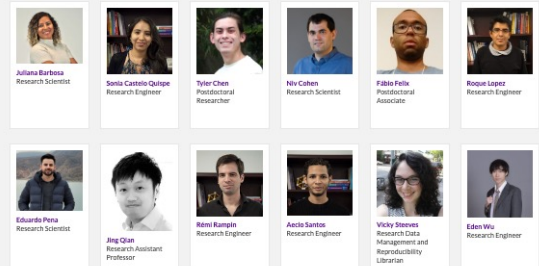
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The VIDA Center

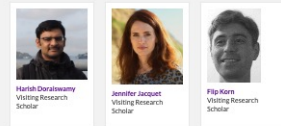
Faculty



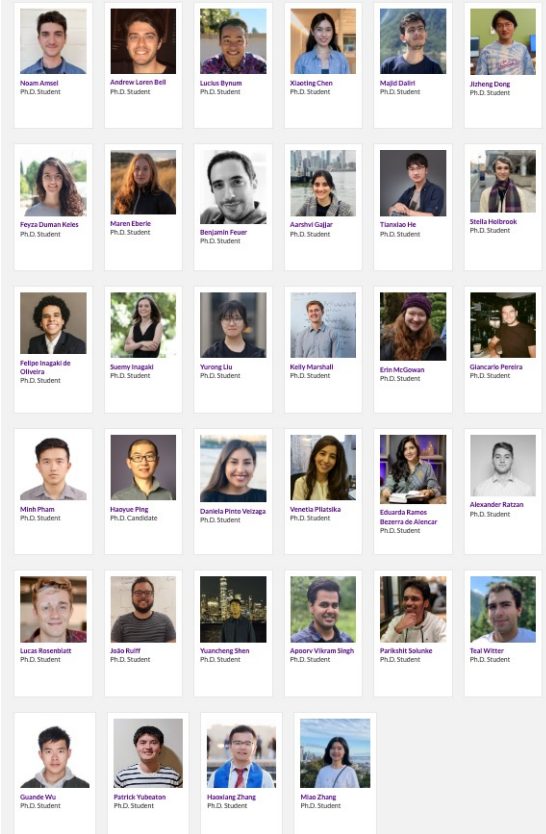
Research Associates



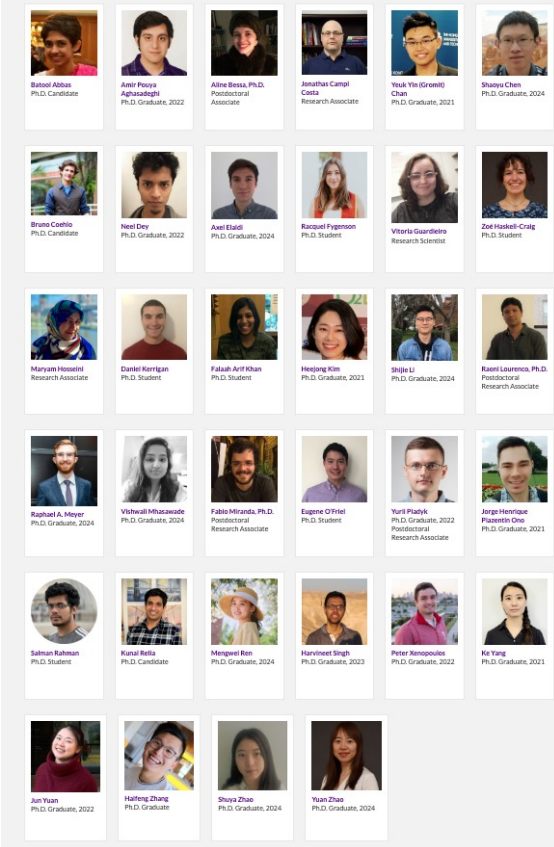
Collaborators and Associated Faculty



Students

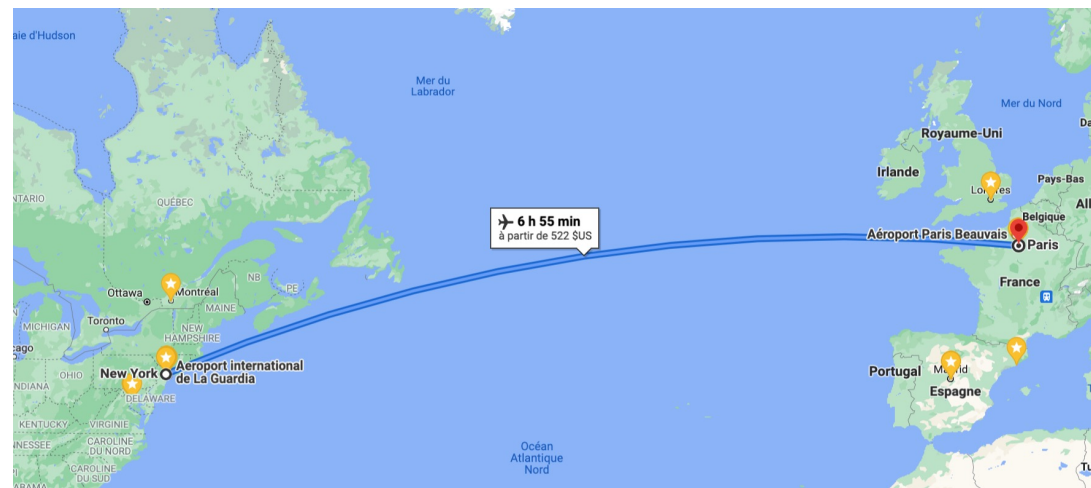


Alumni



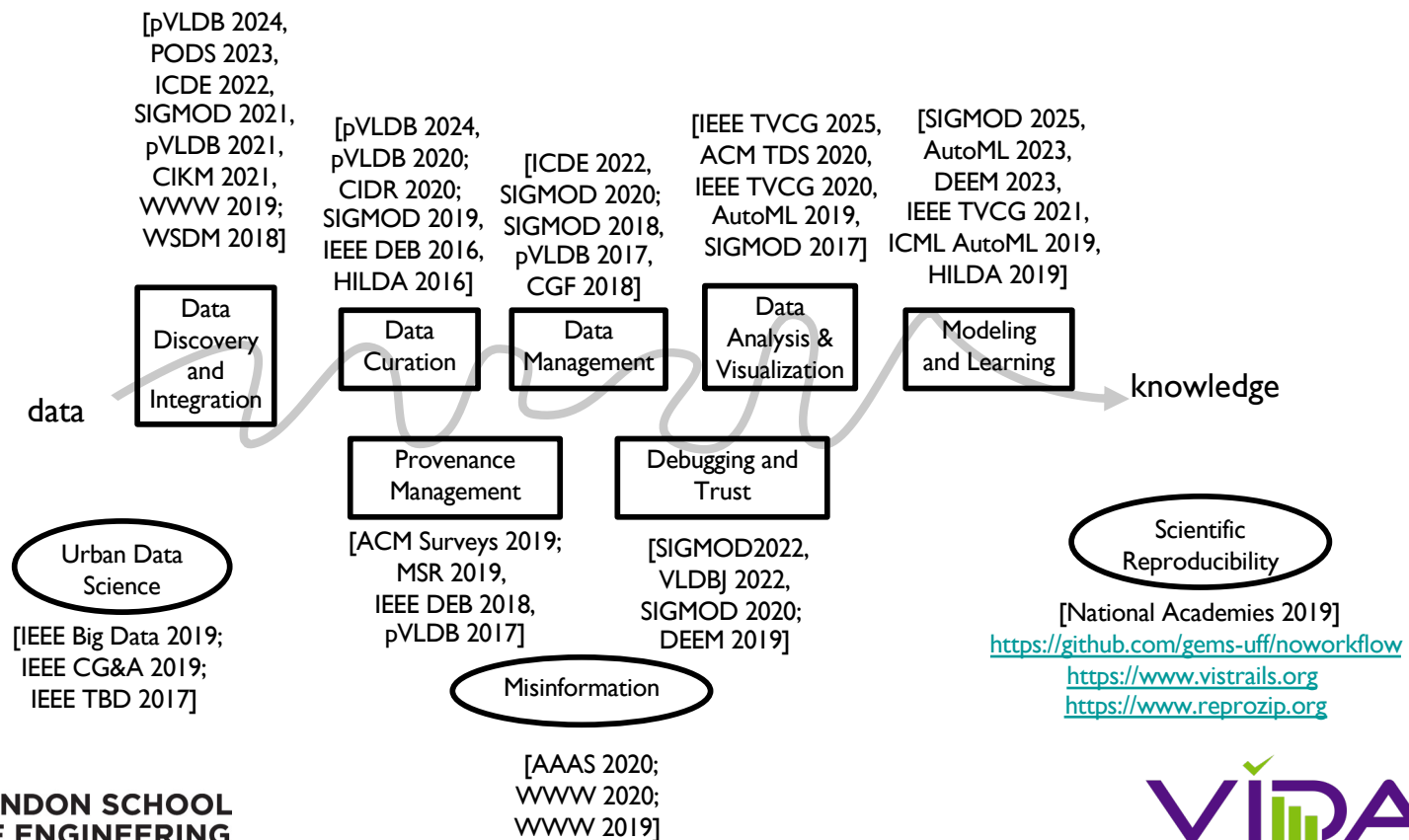
My Journey

- UFC (Brazil) →
SUNY at Stony Brook →
Bell Labs →
OGI/OHSU →
U of Utah →
NYU →
INRIA, France



My Research

Empower a *wide range of users* to explore data and obtain trustworthy insights



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My Research

Address issues at the different stages of the data lifecycle

From concepts and algorithms to deployed systems

- Methods
 - Visualization
 - Data analysis
 - Data management
 - Machine learning
 - Information integration
 - Data mining
 - Provenance management
- Applications
 - Biomedical data integration
 - Urban data analysis
 - Human and Wildlife Trafficking
 - AutoML
 - Data augmentation
 - Data search and discovery
 - Computational reproducibility and transparency

NYU Visualization, Imaging and Data Analysis (VIDA) Center

<http://vida.engineering.nyu.edu>



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Survey

- Fill out <https://forms.gle/MvsG3nPu9BsTqFsA6>



Introductions

Next class in person!



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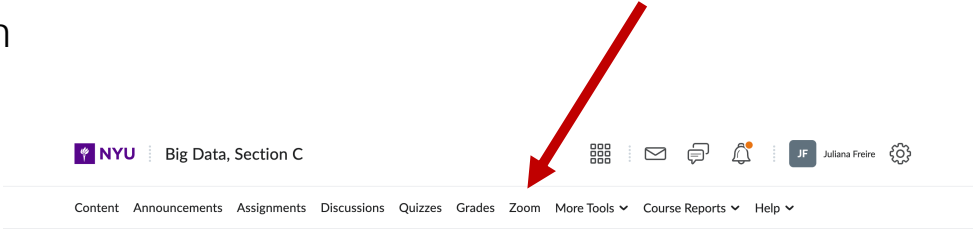
Student Hours in Person and on Zoom

Zoom link: see Brightspace

You must be logged on to zoom using your NYU account!

Juliana Freire

TBD on Zoom and in person



The screenshot shows the Zoom web interface for a course titled "Big Data, Section C". A red arrow points to the "Zoom" link in the top navigation bar. Below the navigation bar, the Zoom logo and a message about the current time zone and language are displayed. The "Upcoming Meetings" tab is selected, showing a list of meetings. The first meeting is "Tomorrow (Recurring) 3:00 PM" with the topic "Big Data, Section C: Juliana's student hours" and meeting ID "950 9631 3255". The other three meetings are "Wed, Feb 8 (Recurring) 3:00 PM", "Wed, Feb 15 (Recurring) 3:00 PM", and "Wed, Feb 22 (Recurring) 3:00 PM", all with the same topic and meeting ID. Each meeting has a "Start" or "Delete" button next to it.

Start Time	Topic	Meeting ID	Start	Delete
Tomorrow (Recurring) 3:00 PM	Big Data, Section C: Juliana's student hours	950 9631 3255	Start	Delete
Wed, Feb 8 (Recurring) 3:00 PM	Big Data, Section C: Juliana's student hours	950 9631 3255		Delete
Wed, Feb 15 (Recurring) 3:00 PM	Big Data, Section C: Juliana's student hours	950 9631 3255		Delete
Wed, Feb 22 (Recurring) 3:00 PM	Big Data, Section C: Juliana's student hours	950 9631 3255		Delete



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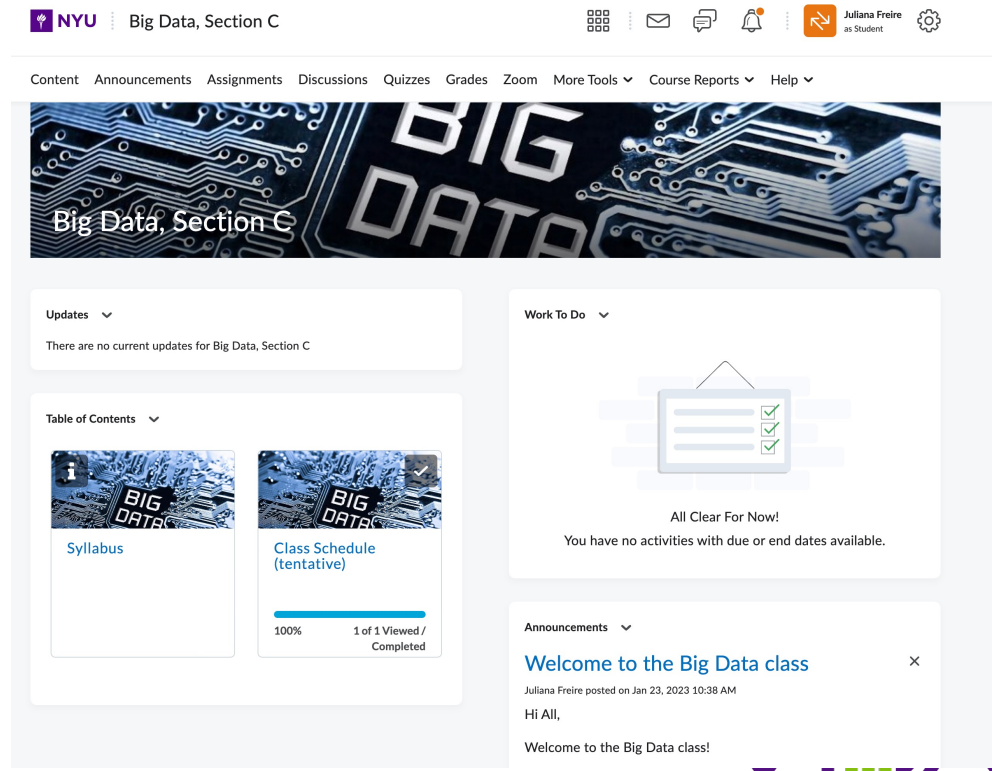
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Course Information: Brightspace

- Course information (e.g., description, zoom links)
- Tentative schedule
- News and announcements
- Reading list
- Assignments
- Project

Check it often!!!



The screenshot shows the Brightspace interface for the course 'Big Data, Section C'. The top navigation bar includes the NYU logo, the course title, and user information for 'Juliana Freire as Student'. Below the navigation bar is a menu with links to Content, Announcements, Assignments, Discussions, Quizzes, Grades, Zoom, More Tools, Course Reports, and Help. The main content area features a header image with the text 'Big Data, Section C'. Below this, there are three main sections: 'Updates' (indicating no current updates), 'Table of Contents' (showing 'Syllabus' and 'Class Schedule (tentative)' with a progress bar at 100%), and 'Work To Do' (indicating 'All Clear For Now!'). An 'Announcements' section at the bottom displays a welcome message from Juliana Freire posted on Jan 23, 2023.

Course Communication: Brightspace

Reach us:
Message Alerts → Email → Compose message

[illegible]

What you will do

- Labs and quizzes (20%)
- Homework and programming assignments (35%)
 - Assignments done *individually – no collaboration (unless otherwise noted)*
 - You will need a computer
 - We will provide access to NYU cloud resources (more details later)
- Final project (40%)
- Class participation (5%) – take into account in class and offline participation
- Become familiar with the rules about academic honesty

No credit for late submissions

http://www.cs.nyu.edu/web/Academic/Graduate/academic_dishonesty.html

Cheating → F in the course

- Our activities will be carried out in an **inclusive and diverse** environment with zero tolerance for discrimination, harassment, or any other form of misconduct

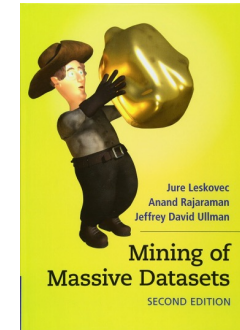
What we will cover

- **Big Data Foundations and Infrastructure:**
 - Background on relational databases
 - Cloud computing
 - Architecture and computing models (e.g., MapReduce, Spark)
 - Distributed and parallel databases
- **Algorithms and Techniques:**
 - Data cleaning and integration
 - Data discovery and augmentation
 - Query processing
 - Data mining and analytics: discovering relationships, finding similar items, frequent itemsets, clustering
- **Applications**
 - Spatial analytics and urban data, reproducibility and trust

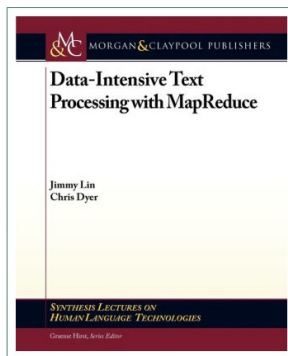
Textbooks

Mining of Massive Data Sets (version 2.1), by Anand Rajaraman, Jure Leskovec and Jeff Ullman. *Free to download!*

<http://www.mmds.org/>



Data-Intensive Text Processing with MapReduce. *Free to download!*
<https://lintool.github.io/MapReduceAlgorithms/MapReduce-book-final.pdf>



We we will also read some
research papers



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Reading Material



Big Data, Section D



Juliana Freire



Content Announcements Assignments Discussions Quizzes Grades Zoom More Tools ▾ Course Reports ▾ Help ▾

+ New Unit



Visible

Add Existing

Create New



Search titles, descriptions



Syllabus

Tentative Schedule

▶ Lectures

Big Data – Week 3: Map Reduce vs. Parallel Databases (09/18/2025)

Description: We will:

- learn about parallel databases and contrast them with Map Reduce
- we will discuss 2 papers (see required reading from 9/11/204)
- do an in-class quiz on the required reading

See slides in [3-paralleldb-vs-hadoop](#)

Required Reading:

- A comparison of approaches to large-scale data analysis. Pavlo et al., SIGMOD 2009. <http://dl.acm.org/citation.cfm?id=1559865>;
<http://dsl.serc.iisc.ernet.in/~course/DBMS/papers/benchmarks-sigmod09.pdf>
- MapReduce: A Flexible Data Processing Tool. Dean and Ghemawat, CACM 2010. <http://cacm.acm.org/magazines/2010/1/55744-mapreduce-a-flexible-data-processing-tool/fulltext>



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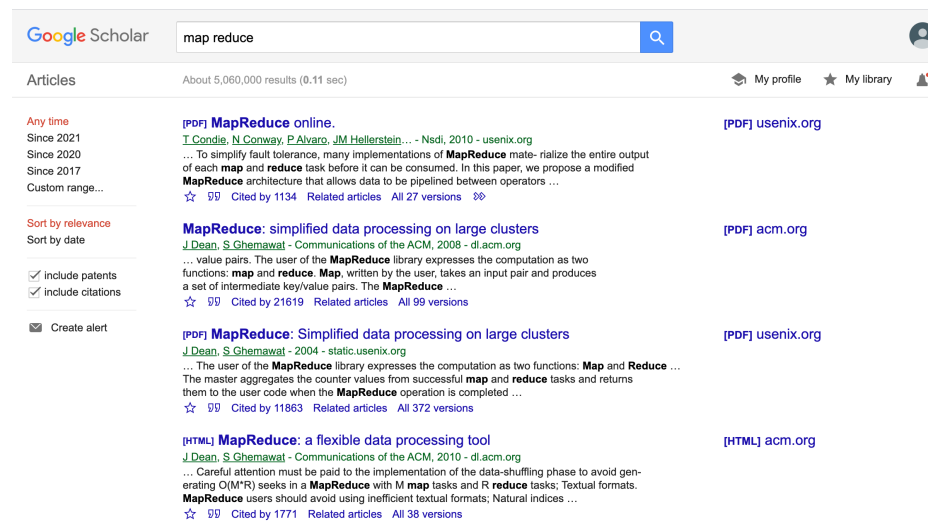
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Reading Material

- You can access papers from within the NYU network
 - If you are off-campus, use the VPN
- You can *easily* find papers using Google Scholar
 - Citations, references



https://scholar.google.com/scholar?hl=en&as_sdt=0%2C33&q=map+reduce&btnG=



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Pre-Requisites

- Knowledge of databases is desirable, but we will cover the basic concepts
- We expect students to have taken a course and/or be familiar with
 - Algorithms and data structures
 - Programming – we will use Python in the course

Computing Resources

- You will use the NYU HPC cluster for homework assignments and project
- All registered students should have access to Dataproc

If you don't let me know!

Big Data Management: Overview

Juliana Freire

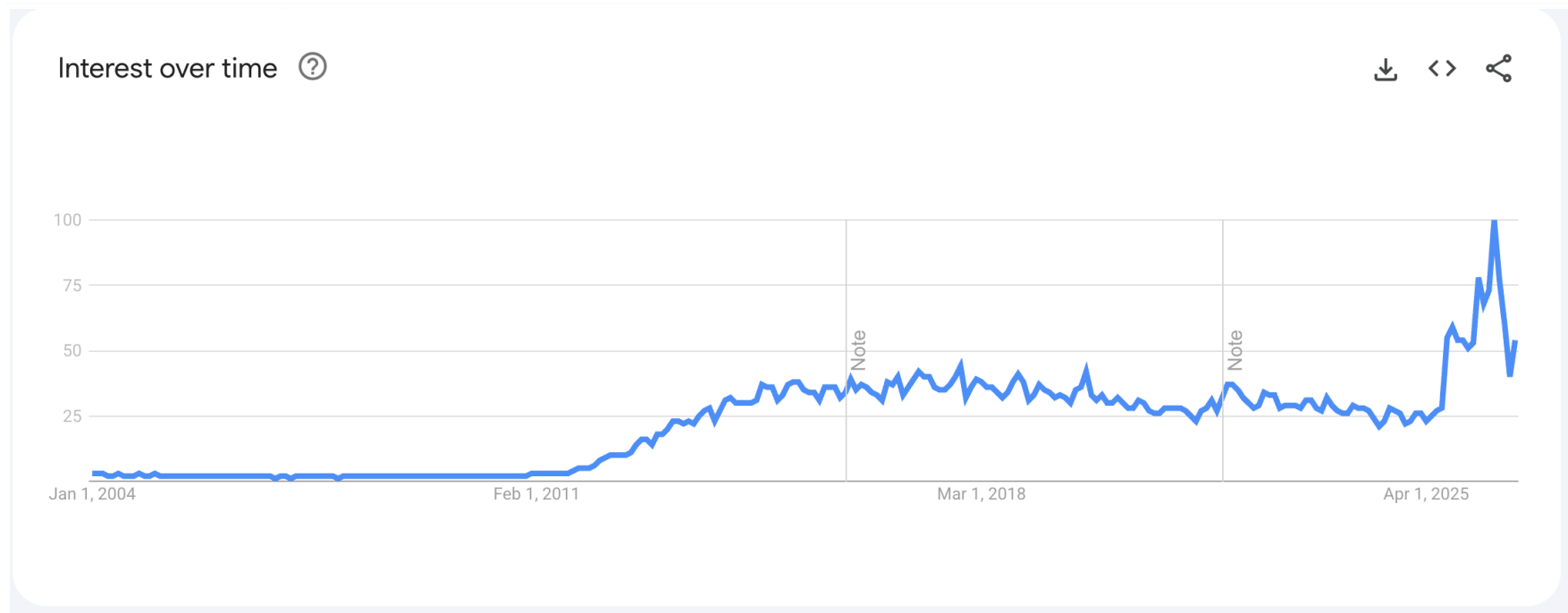


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Google Trends: Big Data



<https://www.google.com/trends/explore?date=all&q=big%20data>

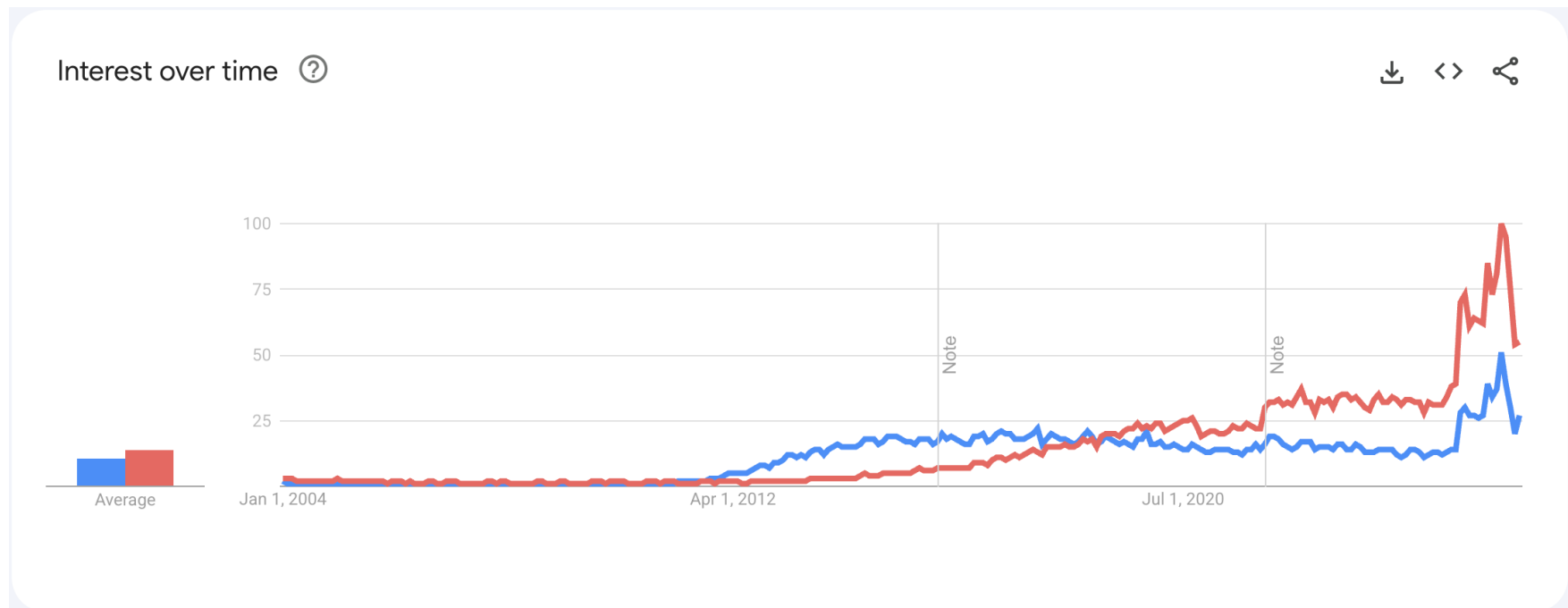


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Google Trends: Big Data vs. Data Science



<https://trends.google.com/trends/explore?date=all&q=big%20data,data%20science>

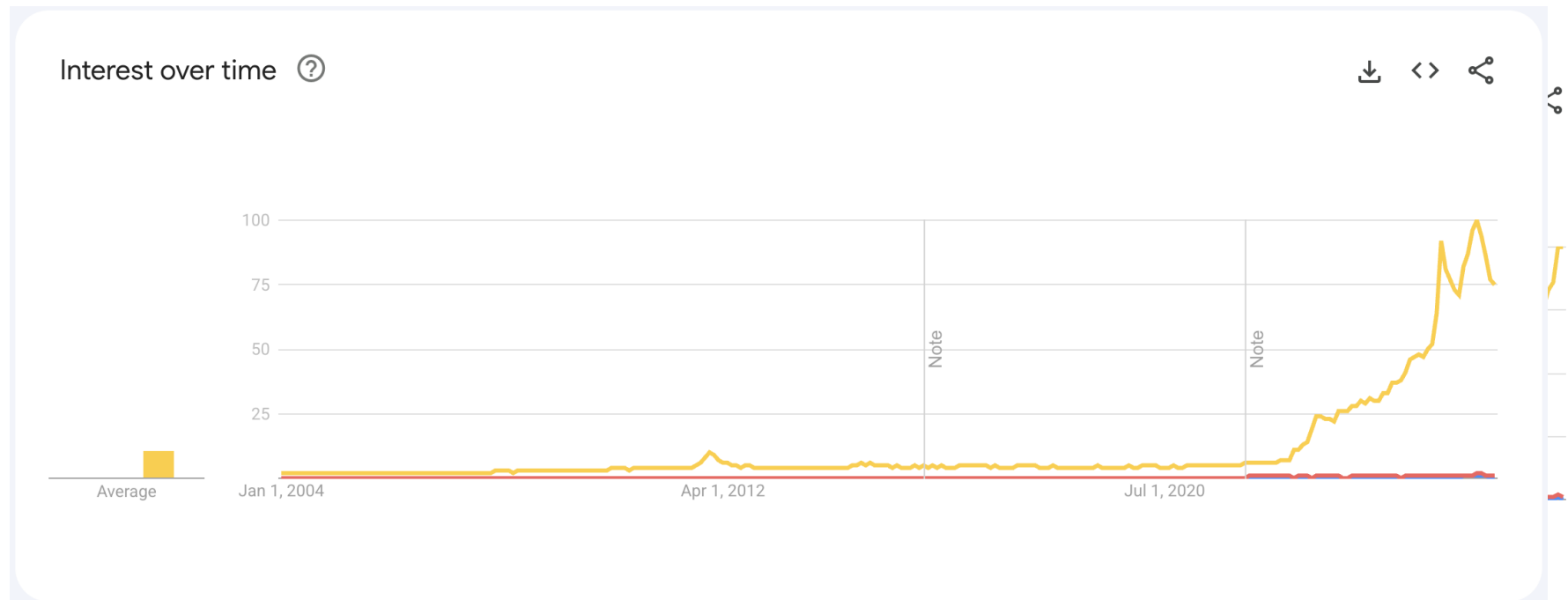


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Google Trends: Big Data vs. AI



<https://trends.google.com/trends/explore?date=all&q=big%20data,data%20science,AI>



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Is big data less important today?

There is no AI without Big Data...



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What is Big Data?

The four V's of big data: Volume, Variety, Velocity, Veracity

- *Volume*: petabyte-scale collections or lots of (not necessarily big) data sets
- *Variety*: does not fit neatly in an existing tool
 - Data sets that need to be cleaned, processed and integrated
 - E.g., Twitter, news, customer transactions, web logs, videos
- *Velocity*: needs to be processed quickly – streaming data
- *Veracity*: different levels of accuracy and quality

More V's: Variability, Value, Vulnerability, ...

<http://www.ibmbigdatahub.com/infographic/four-vs-big-data>

<https://tdwi.org/articles/2017/02/08/10-vs-of-big-data.aspx>



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Big Data in Astronomy

Volume

Table 1

Excel | CSV

Data volumes of different sky survey projects.

Sky Survey Projects	Data Volume
DPOSS (The Palomar Digital Sky Survey)	3 TB
2MASS (The Two Micron All-Sky Survey)	10 TB
GBT (Green Bank Telescope)	20 PB
GALEX (The Galaxy Evolution Explorer)	30 TB
SDSS (The Sloan Digital Sky Survey)	40 TB
SkyMapper Southern Sky Survey	500 TB
PanSTARRS (The Panoramic Survey Telescope and Rapid Response System)	~ 40 PB expected
LSST (The Large Synoptic Survey Telescope)	~ 200 PB expected
SKA (The Square Kilometer Array)	~ 4.6 EB expected

Variety

- Images, spectra, time-series data, and simulation data
- Data saved in catalogues or databases
- Data from different telescopes or projects have their own formats -- difficult to integrate data from various sources

Velocity

- LSST generates one SDSS (116TB) each night for 10 years.
- Expects to find 1000 new supernovae each night for 10 years -- 10-100 thousand alerts



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<https://bit.ly/36wq9Q0>



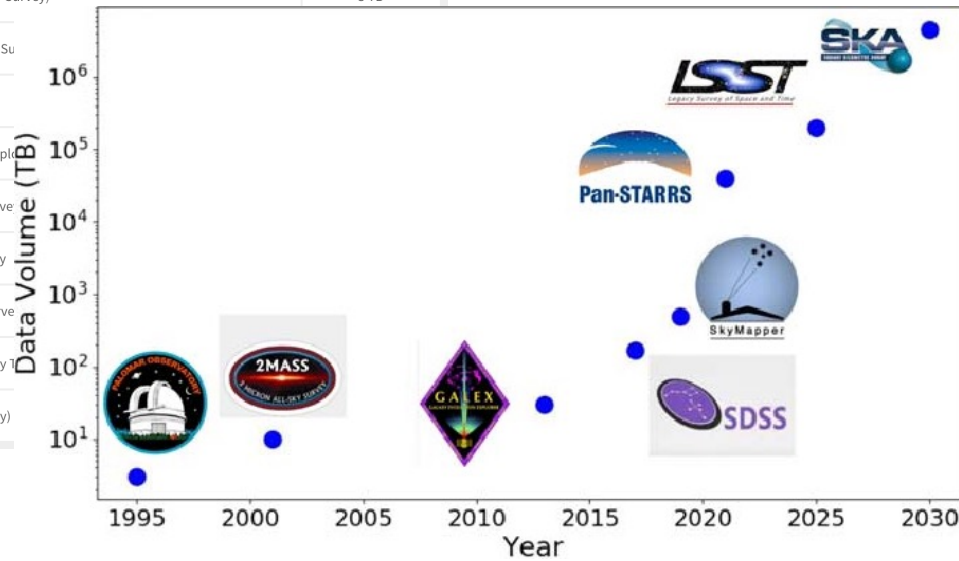
Big Data in Astronomy

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SKA (The Square Kilometer Array)	



Variety

- Images, spectra, time-series data, and simulation data
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- Data from different telescopes

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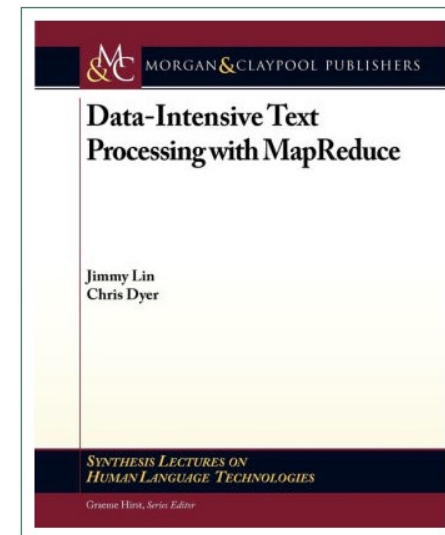
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Poll: Volume

Share examples of problems/applications that require or can benefit from *large data volumes*

Volume

“How much data are we talking about? A few examples: **Google** grew from processing 100 TB of data a day with MapReduce in 2004 [...] to processing **20 PB a day** with MapReduce in 2008 [...]. In April 2009, a [blog post](#) was written about **eBay**’s two enormous data warehouses: one with **2 petabytes of user data**, and the other with **6.5 petabytes of user data spanning 170 trillion records and growing by 150 billion new records per day**. Shortly thereafter, **Facebook** [revealed](#) similarly impressive numbers, boasting of **2.5 petabytes of user data, growing at about 15 terabytes per day**.”



[Lin and Dyer, 2010]



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Volume: More recent estimates

Google search: In 2017 it was reported that Google “processes over 40,000 search queries every second on average, which translates to over 3.5 billion searches per day and 1.2 trillion searches per year worldwide.”

YouTube, Gmail, Google Maps....

“Back in 2013, Randall Munroe of xkcd tried to calculate how much data google stores and came out with 10 Exabytes. That’s equivalent to 10,000 Petabytes and 10,000,000 Terabytes.”

<https://lewisdgavin.medium.com/googles-data-footprint-will-blow-your-mind-2237cf8e0d4>

Claude (2025): Google processes over 8.5 billion searches daily,
indexing hundreds of billions of web pages



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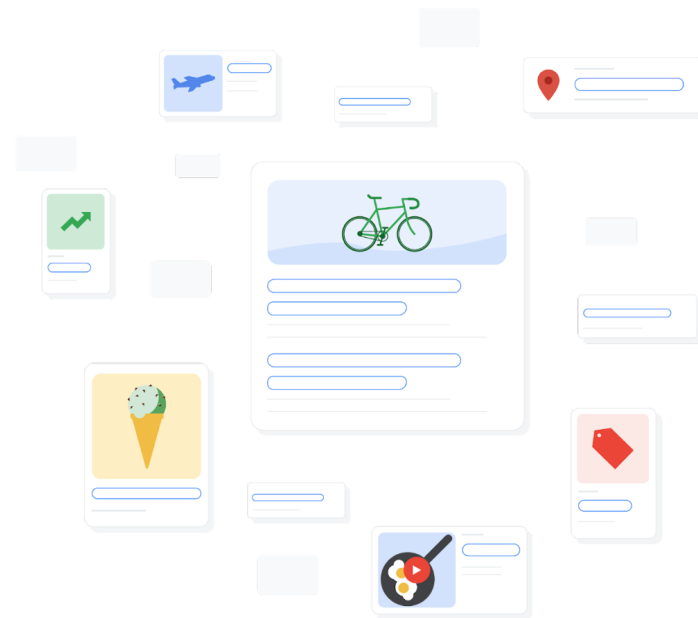
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Google Search Today

How Google Search organizes information

When you Search, Google looks through hundreds of billions of webpages and other content stored in our Search index to find helpful information — more information than all of the libraries of the world.



Volume

- Large data sets or (too) many data sets
- **Challenge:** how to store and process data efficiently
- **Approaches:**
 - Relational databases
 - Parallel databases
 - Parallel/distributed/cloud architectures
- Many trade-offs: cost, ease of use, support of interactivity
- No one-size-fits-all: different tasks/applications require different approaches
- We will learn about distributed architectures, their pros and cons

Poll: Variety

Share examples of problems/applications that require or can benefit from a *large number of data types/sources*

Variety

- Data comes in different forms
 - Format (structured, unstructured, spatial, temporal); content (text, images, videos, sensors); domain (e-commerce, health, scientific experiments, traffic, weather)
- Data comes from multiple sources
- Challenge: how to process/analyze and combine these data?
- Approaches:
 - Data integration/harmonization, Data lakes, Data explanation
- We will learn about methods that support data discovery and integration

Veracity

- Data can be noisy, incomplete, and biased
- Challenge: how to *clean* data? How to derive robust insights from imperfect data?
- Approaches:
 - Data profiling and cleaning
 - Data provenance
 - Identify and mitigate bias
 - Debugging data+computation
- We will learn about data profiling, cleaning, debugging, and provenance

Big Data: What is the **Big** deal?

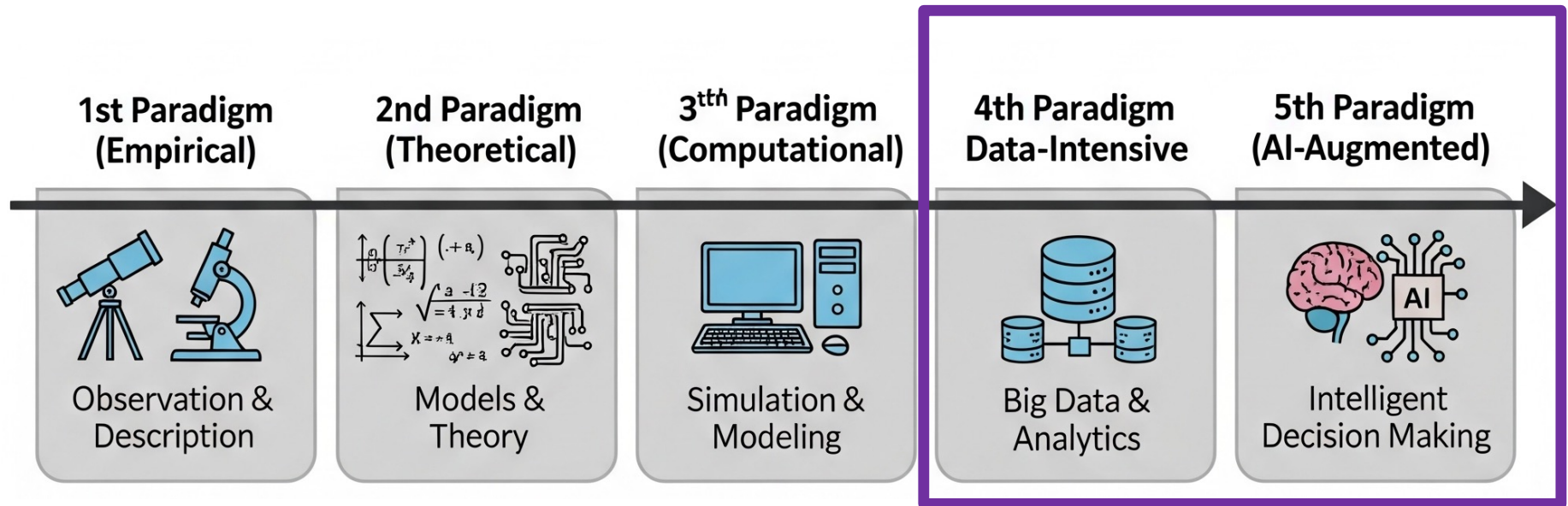


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Science Paradigms and New Data Challenges



Generated by Gemini 2.5 flash



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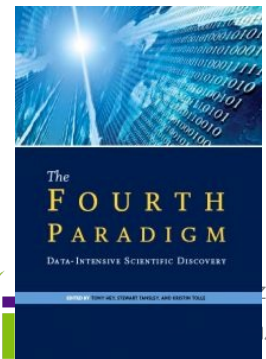
Big Data: What is the **Big** deal?

- Every scientific domain has moved toward data-driven exploration; this has led to great advances and discoveries
 - Urban computing research enabled by open city data
 - Social sciences research enabled by social media data
 - Citations in Google Scholar over time
 - Facebook: 2015: 3.1M ; 2016: 4.8M; 2017: 5.3M; 2020: 6.4M; 2021: 6.69M; 2023: 7.2M
 - Twitter -- 2015: 3.4M; 2016: 5.8M; 2017: 6.39M; 2020: 7.4M; 2021: 7.61M; 2023: 7.7M
 - Petabytes of data generated each day, e.g., Australian radio telescopes, Large Hadron Collider, Sloan Sky Survey, genomes, climate data,



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<https://er.educause.edu/articles/2012/6/scientific-research-how-many-paradigms>

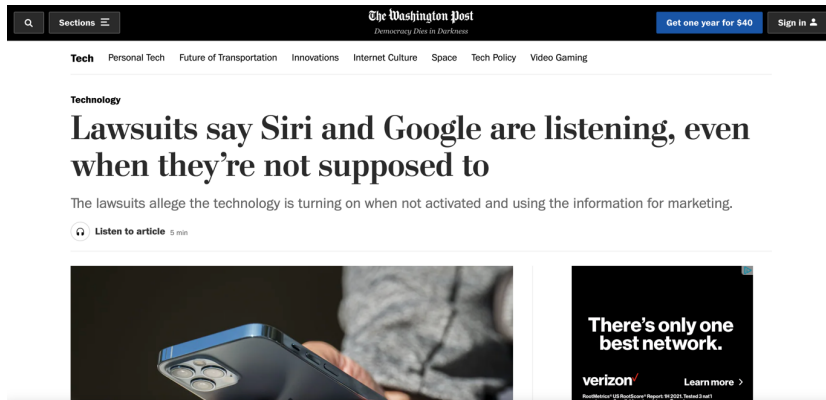


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Big Data: What is the **Big** deal?

- Companies are capitalizing on data
- Amazon online shopping – many users consuming and producing data
- Google: Gmail, YouTube, Ads, Google Home, ...
- Apple: iPhone -- location, computer use, conversations,

<https://www.washingtonpost.com/technology/2021/09/02/apple-siri-lawsuit-privacy/>



Amazon charged with abusing EU competition rules

© 10 November 2020



The European Commission has charged Amazon with abusing its dominant position in online retail to gain an unfair advantage over competitors.

It said Amazon had used data on third-party sellers that use its marketplace to boost sales of its own-label goods.

<https://www.bbc.com/news/business-54887650>



Big Data: What is the **Big** deal?

- Government agencies use data to operate efficiently, make policies, and informed decisions
- E.g., data.gov -- 224,669 datasets; NYC Open Data -- 2,000+ data sets



Big Data: What is the **Big** deal?

Perfect storm:

Computing is free

Storage is free

Data are abundant

Many success stories!



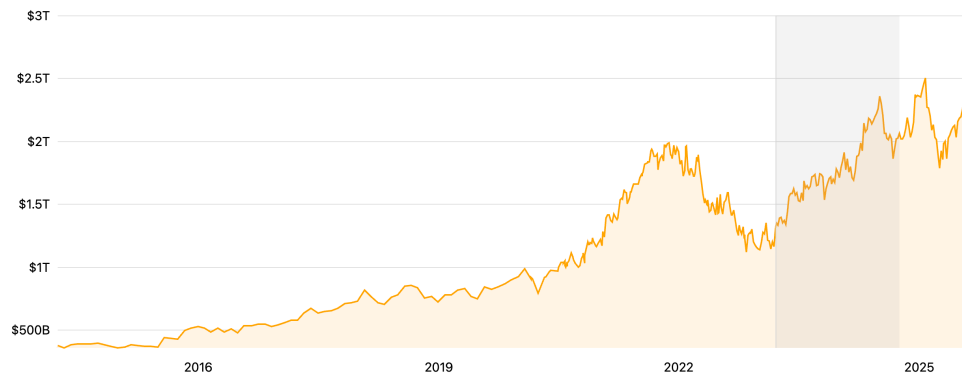
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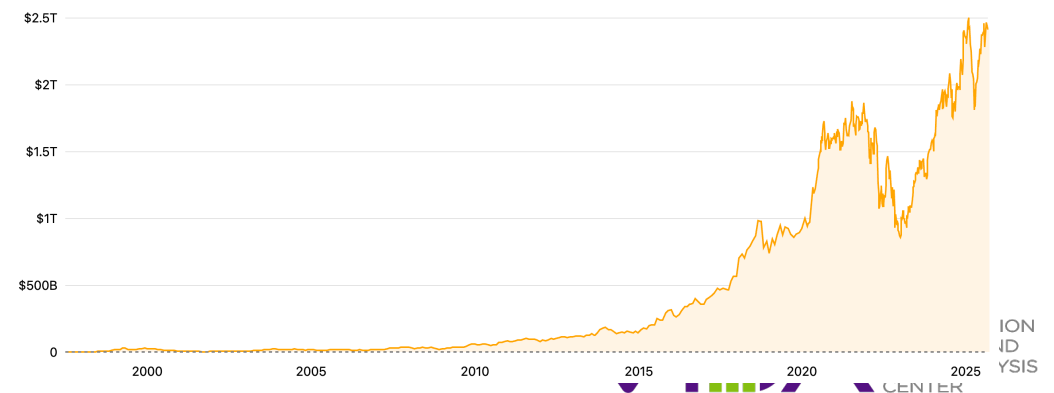
Big Data: Success in Industry

Market cap history of Alphabet (Google) from 2014 to 2025



This has changed society! For better and worse...

Market cap history of Amazon from 1997 to 2025



Big Data Propagation: Fake News

The Fix • Analysis

A new study suggests fake news might have won Donald Trump the 2016 election



Screenshot from 2016 Donald Trump campaign ad, "Dangerous."

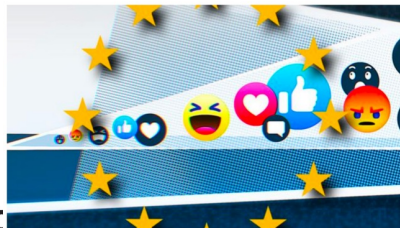
BBC Trending

European elections: How disinformation spread in Facebook groups

By Marianna Spring and Lucy Webster
BBC Newsnight

30 May 2019

f t y e Share



Social media groups which encouraged support for UK political parties in the build up to the European elections were targets for fake news and polarising

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World Europe US Americas Asia Australia Middle East Africa Inequality Global development

Brazil

This article is more than 3 months old

Brazil police target key Bolsonaro supporters in fake news raids

- Carlos Bolsonaro, president's son, reportedly a key member
- Phones and computers seized from hardcore Bolsonaroists



Tom Phillips in Rio de Janeiro

Wed 27 May 2020 11:41 EDT

Latest Posts

How Covid-19 myths are merging with the QAnon conspiracy theory
Online and in real-life demonstrations, two viral conspiracy theories are increasingly coming together.

3 September 2020

Coronavirus: Health worker investigated by employer after posting conspiracy video

28 August 2020

Coronavirus: The viral rumours that were completely wrong

6 August 2020

Black Lives Matter: Can viral videos stop police brutality?

6 July 2020

Subscribe to our podcast



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Poll: What Big Data “Vs” are related to Fake News?

- a) Variety
- b) Volume
- c) Veracity
- d) Velocity



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Big Data: Success in Science

DeepMind AI solves a half-century-old protein problem

In November it was revealed that an AI lab based in London had solved a mystery that had puzzled experts for 50 years, by predicting the 3D shape of proteins from their sequence of amino acids. Proteins, essentially the building blocks of life, are

<https://www.newsweek.com>

Big Data is Transforming How Astronomers Make Discoveries

The next game-changer is likely lurking in the data we already have—but it will take scientists years to uncover it



<https://www.smithsonianmag.com/science-nature/next-big-discovery-astronomy-scientists-probably-found-it-years-ago-they-dont-know-it-yet-180969073/>

AI is powered
by Big Data!

COVID-19

Climate change

...

Enormous Coral Reef Taller than Empire State Building Discovered

"...The state of our knowledge about what's in the ocean has long been so limited. Thanks to new technologies that work as our eyes, ears and hands in the deep ocean, we have the capacity to explore like never before..."

<https://www.newsweek.com/incredible-scientific-discoveries-2020-1557134>



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Big Data: Success in Cities

- Real-time bus arrival predictions
 - 94% reported increased or greatly increased satisfaction with public transit



Benefit residents

Big Data: Success in Cities

- Illegal conversions in NYC
 - DS team (1) integrated data from 19 different agencies that provided indication of issues in buildings; (2) crossed with fire data; (3) Created a prediction model
 - Hit rate for inspections went from 13% to 70%



Todd Heisler/The New York Times

Michael Flowers, right, oversees a small group of tech-savvy and civic-minded statisticians working across from City Hall.

Make cities more efficient



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Big Data: Success in Cities

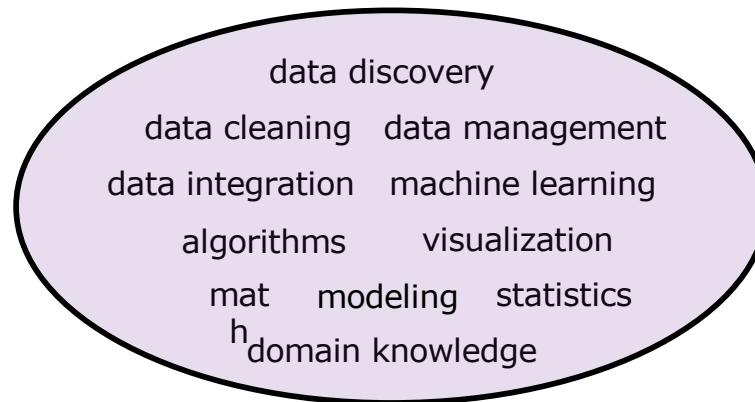
Impact policy

- Foreclosures and crime
 - Neighborhoods with concentrated foreclosures see an uptick in crime for each foreclosure notice issued
 - NYPD updated its policing strategies



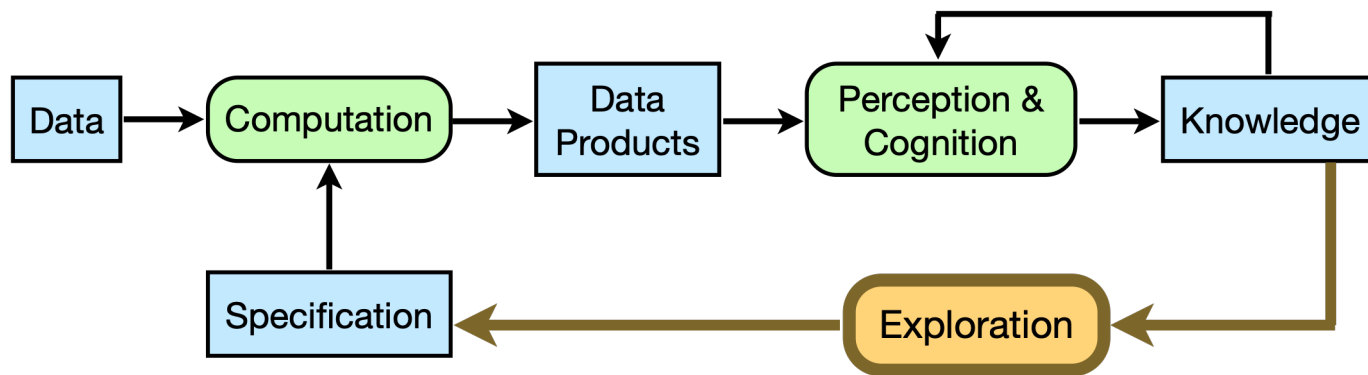
Big Data Exploration: Challenges

- Data are vast and produced at unprecedented rates
 - Sources are broad, varied, and unreliable
- Computational processes are required to extract insight
 - But they hard to assemble and require expertise in a wide range of topics



Big Data Exploration: Challenges

- Data are vast and produced at unprecedented rates
 - Sources are broad, varied, and unreliable
- Computational processes are required to extract insight
 - But they hard to assemble
- Exploratory tasks are inherently iterative as one tests and formulates hypotheses



[Modified from Van Wijk, Vis 2005]

Big Data Exploration: Challenges

- After many steps...

"An analysis has 30 different steps. It is tempting to just do this then that and then this. You have no idea in which ways you are wrong and what data is wrong" [Kandel et al., VAST 2012]

- It is easy to get lost and not remember how a result was derived
- Processes can break or misbehave in unforeseen ways
- Results can be hard to understand, interpret and trust



Incorrect conclusions can lead to bad decisions

Need provenance and debugging

Big Data: What is the bottleneck?

Computing is free

Storage is free

Data are abundant

The bottleneck lies with people!

The McKinsey Report on Big Data

Data have swept into every industry and business function and are now an important factor of production, alongside labor and capital. We estimate that, by 2009, nearly all sectors in the US economy had at least an average of 200 terabytes of stored data (twice the size of US retailer Wal-Mart's data warehouse in 1999) per company with more than 1,000 employees...

The use of big data will become a key basis of competition and growth for individual firms.

*There will be a shortage of **talent** necessary for organizations to take advantage of big data. By 2018, the United States alone could face a shortage of 140,000 to 190,000 people with deep analytical skills as well as 1.5 million managers and analysts with the know-how to use the analysis of big data to make effective decisions.*



What has happened?

“...data scientist has been ranked the #1 Job in America for three years running”
[Glassdoor 2018]

“businesses across the nation need 151,717 more data scientists right now ... shortage of people with software development skills is greater” [LinkedIn 2018]

“data engineers are more in demand than data scientists” [LinkedIn 2023]

<https://www.nasdaq.com/article/the-best-jobs-in-america-in-2018-and-where-to-find-them-cm1012158>

<https://www.nasdaq.com/articles/best-jobs-america-2018-and-where-find-them-2018-08-23>

<https://economicgraph.linkedin.com/resources/linkedin-workforce-report-august-2018>

<https://www.linkedin.com/pulse/data-scientist-vs-engineer-reloaded-michael-spencer>

Some Big Data Problems I have worked on



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Using Big Data to Fight Crime

- *Develop software that advances online search capabilities ... invent better methods for interacting with and sharing information, so users can quickly and thoroughly organize and search subsets of information relevant to their individual interests ... improve content discovery, information extraction, information retrieval, user collaboration and other key search functions.*

The image shows two news articles side-by-side. The left article is from Forbes, dated February 8, 2015, titled "NEW SEARCH ENGINE EXPOSES THE 'DARK WEB'". It mentions that Memex, a powerful new search tool developed by DARPA, is launched by DARPA. The right article is from Scientific American, dated February 8, 2015, titled "Human Traffickers Caught on Hidden Internet". It mentions that a new set of search tools called Memex, developed by DARPA, peers into the "deep Web" to reveal illegal activity.

Forbes / Security

APR 10, 2015 @ 06:39 AM 25,764 VIEWS

Memex In Action: Watch DARPA Artificial Intelligence Search For Crime On The 'Dark Web'

NEW SEARCH ENGINE EXPOSES THE "DARK WEB"

Memex, a powerful new search tool that goes beyond the realm of Google, Yahoo, and Bing, is launched by DARPA

2015 FEB 08 COMMENTS 18 FACEBOOK 4.9K TWITTER STUMBLE MORE

This week on 60 Minutes, Lesley Stahl and producer Shachar Bar-On got an early look at Memex, a powerful new search engine developed by DARPA, the U.S. military's Advanced Research Projects Agency.

SCIENTIFIC AMERICAN

English Cart

THE SCIENCES MIND HEALTH TECH SUSTAINABILITY EDUCATION VIDEO PODCASTS BLOGS S

TECH

Human Traffickers Caught on Hidden Internet

A new set of search tools called Memex, developed by DARPA, peers into the "deep Web" to reveal illegal activity

By Larry Greenemeier on February 8, 2015

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DARPA

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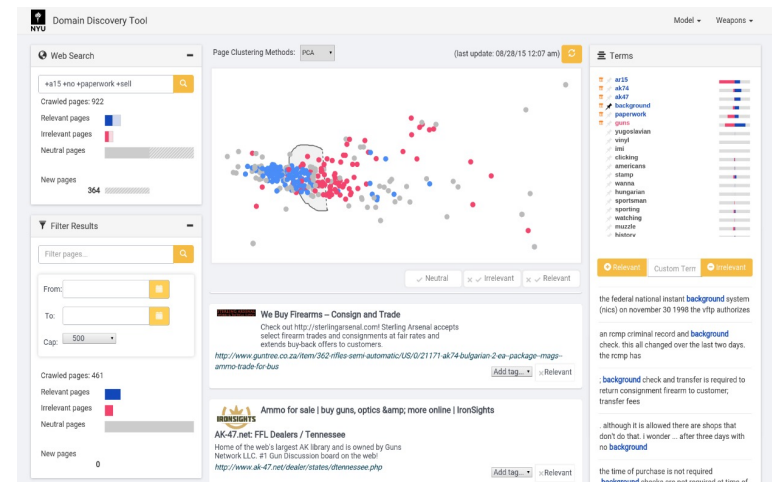
Memex: Domain-Specific Data Discovery on the Web

- Goal: enable domain experts to find information on the Web
- Challenges:
 - Too much information on the Web, finding relevant information is like finding a needle in a haystack
 - Difficult to specify information being sought
 - Information changes constantly, sites/pages disappear and are created constantly

Domain-Specific Data Discovery on the Web

Goals:

- Translate a user's conceptual view of a domain/information need into a computational model
- Combine techniques from *machine learning*, *information retrieval* and *visualization*
- Guide users in the model construction process
- Develop scalable methods to discover and monitor *relevant* information on the Web



Data Discovery: Research Challenges

- Guide the user in the classifier construction process
 - Further reduce the need for user input: use active learning to rank pages wrt importance, i.e., ask user to label pages that will lead to bigger improvements in the quality of the classifier
 - Provide feedback to the user on the quality of the classifier
 - **Help user understand the classifier**
- Guide the user in the process of classifier refinement
 - Visual analytics techniques to help user understand how classifiers work by analyzing their output, e.g., compare different crawl snapshots
 - Visual representations to summarize a corpus and its evolution
- Scalable and *secure* information discovery through focused crawling: surface, deep and dark Web
- Information extraction and analytics

Domain Discovery Tool (DDT)

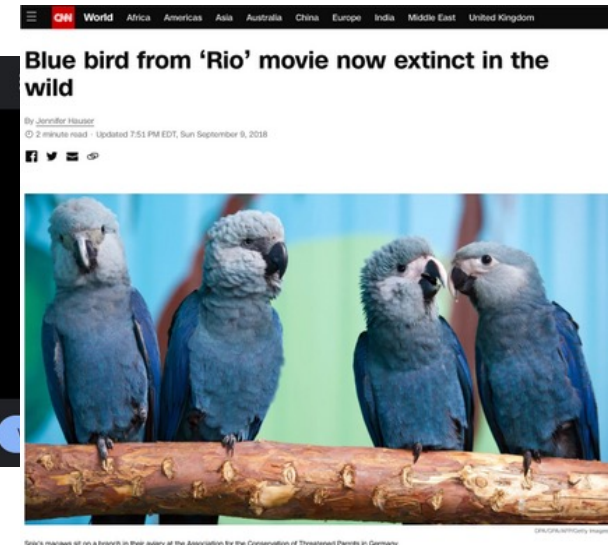
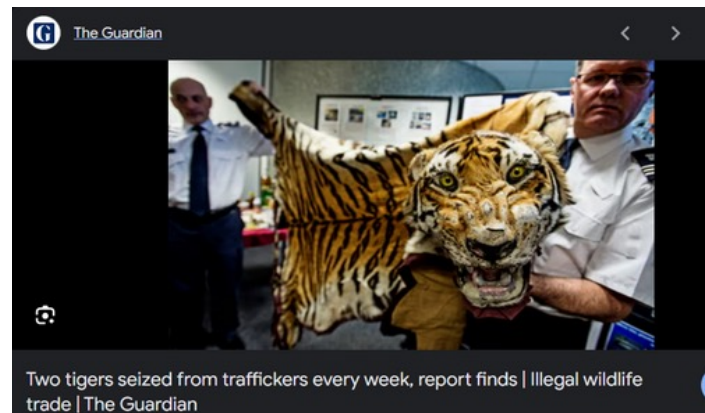
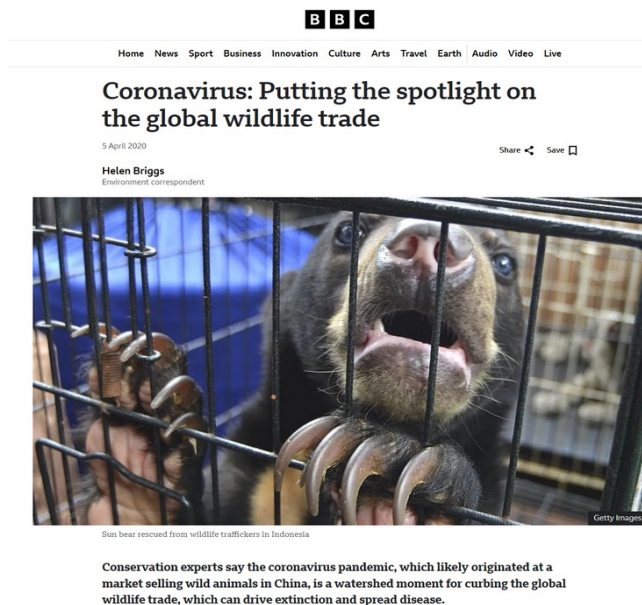
- Enables domain experts to explore a domain
- Provides visual summaries of results returned by search engines
 - Helps users formulate new queries
- Users interactively provide feedback wrt the relevance of pages
- Incrementally build a learning classifier to recognize relevant pages
- Time consuming...

https://github.com/ViDA-NYU/domain_discovery_tool

[ACM KDD IDEA 2016]

Wildlife Trafficking

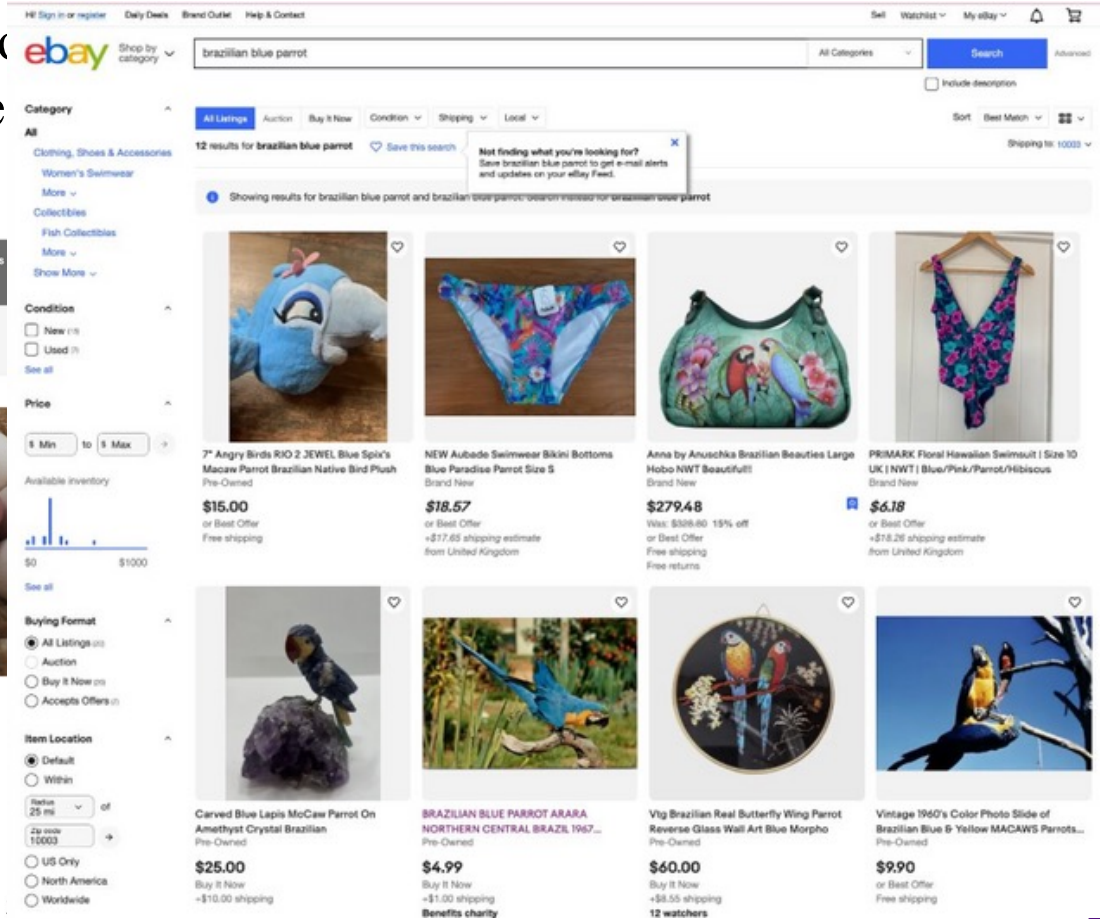
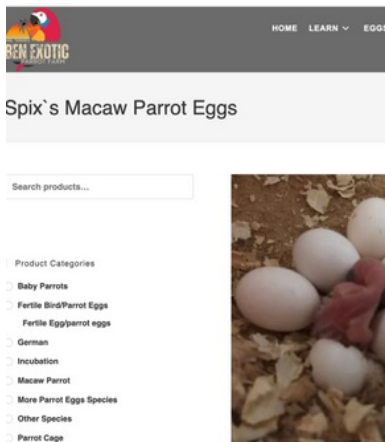
Critical global issue, significantly impacting biodiversity, ecological stability, and public health.



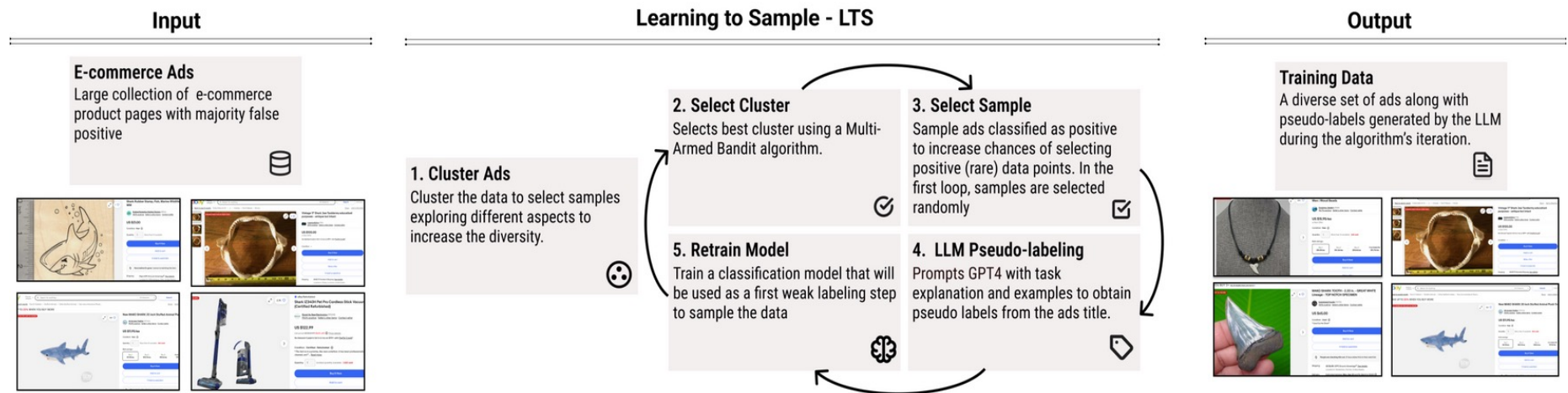
Online Wildlife Trafficking

Opportunity to discover (low volume) that enable disrupt it

Patterns, trade interventions to



Automatic Creation of Triage Classifiers



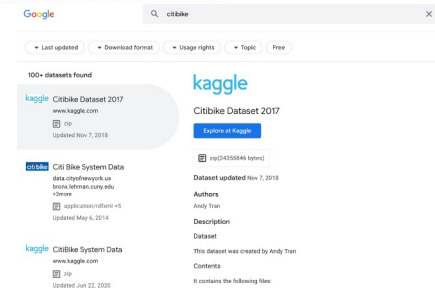
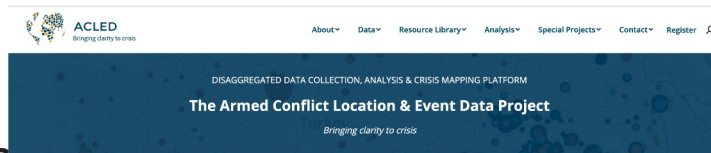
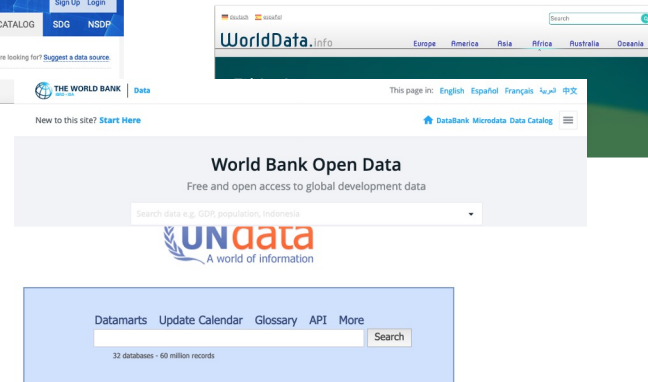
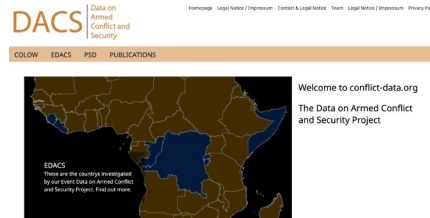
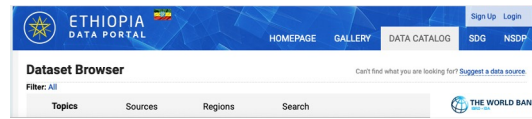
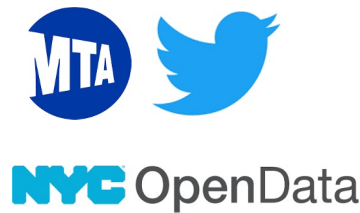
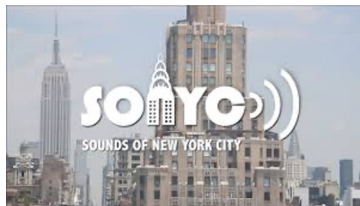
Leverage LLMs to label data!

Focused Crawling: Research

- ACHE: <https://github.com/ViDA-NYU/ache>
- Understanding crawler coverage and effectiveness
- Domain monitoring and expansion [WWW 2016, WSDM 2018]
- Object-focus crawling: many search tasks seek well-defined objects, e.g., forum threads/posts, weapons for sale, escort ads
 - Domain experts often provide a small number of seeds/sites, e.g., a set of interesting forums or suspicious stores
 - Problem: How to automatically learn and refine a site classifier?

Dataset Discovery and Search

- Increased ability to capture and store data -- a very large number of datasets
- Opportunity to answer important questions: enrich analytics, explain data, and improve machine learning models



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How Many Datasets?

Google Dataset Search indexes
28 million datasets from more
than 3,700 sites
[Benjelloun et al., ISWC 2020]

Google Dataset Search indexes
45 million datasets
[Noy 2023]

200-300 million datasets
on the Web
[Noy 2023]



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Data Abundance: Opportunities

- Test and generate hypotheses
- Create and refine machine learning models
- Enrich data-driven analyses, explain features and findings
- Help answer scientific and societal questions
 - Explore and compare different NYC neighborhoods wrt crime, transportation, noise, taxi service, etc.
 - Can we predict CTNNB1 mutation status from proteomics data only?

Data Discovery: Challenges

- Data discovery is a major bottleneck in data-driven exploration
 - Data scientists spend 25% of their time on data discovery [Lyft 2019]



How can we discover datasets on the Web
and make it easy to search/query dataset
collections?

- Finding data is difficult



Huge gap between data
producers and
consumers

Dataset Repositories, Catalogs, and Search Engines

- Datasets are spread over multiple sites: organize datasets (or their metadata) in centralized repositories

The collage illustrates various dataset repositories and search engines:

- NYC OpenData:** A screenshot of the NYC OpenData website showing search results for "bike". It lists categories like Business, City Government, Education, Environment, and Health. A specific result for "Citi Bike System Data" is highlighted, showing it was updated on September 10, 2018, and has 15,530 views.
- Harvard Dataverse:** A screenshot of the Harvard Dataverse website showing search results for "covid". It displays metrics such as 35,629,658 downloads and a search bar with "covid" entered.
- Kaggle:** A screenshot of the Kaggle website showing search results for "nyc precipitation". It lists datasets like "Weather data in New York City - 2016" and "Climate Change Projections and Impacts for New York State".
- Schema.org:** A screenshot of the Schema.org website showing a "Dataset" type. It defines a Dataset as "A body of structured information describing some topic(s) of interest."
- Smoking Data of hospitalized Covid-19 patients:** A snippet of a dataset entry from Harvard Dataverse, dated Feb 14, 2022, describing data related to 1040 patients with Covid-19 admitted to hospitals in Iran.

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Dataset Repositories, Catalogs, and Search Engines

NYC OpenData

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🔍 citi bike

Keyword queries

20 Results Sort by Most Relevant ▾

Citi Bike System Data NYC BigApps	↗ External Link
Data to solve Citi Bike's Big Idea.	Updated September 10, 2018
Views 18,828	
Tags transportation, bigapps, city government, big apps, citibike, and 1 more	
New York City Bike Routes Transportation	🗺 Map
Locations of bike lanes and routes throughout the City	Updated February 25, 2022
Views 28,011	
Tags 2018od4a-video, nyc bike routes, bike routes	
Citi Bike Live Station Feed (JSON) NYC BigApps	↗ External Link
Data to solve Citi Bike's BigIdea.	Updated September 17, 2018
Views 9,039	
Tags citibike, big apps, city government, transportation, bigapps, and 2 more	
Bike Share Inspections Transportation	📄 Dataset
DOT field inspectors conduct inspections and issue tickets in real-time to the Bike Share system operator according to service level agreements. Stations: docks, bikes, and bikes are available.	Updated February 11, 2022
Views 1,844	
Tags share, inspection, ticket, citi bike, citibike, and 3 more	API Docs

Textual
snippets

Do for datasets what search
engines have done for
documents



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Querying Dataset Collections

- Difficult to express information needs using keyword-based queries and to assess dataset relevance based on textual snippets

NYC OpenData

Home Data About Learn All

Q citi bike

Categories

Business

City Government

Education

Environment

Health

Show All...

View Types

Data Lens pages

Datasets

External Datasets

Files and Documents

Filtered Views

Maps

Data Collection

20 Results

Citi Bike System Data

NYC BigApps

Data to solve Citi Bike's Big Idea.

Tags transportation, bigapps, city government, big apps, citibike, and 1 more

New York City Bike Routes

Transportation

Locations of bike lanes and routes throughout the City

Tags 2018od4a-video, nyc bike routes, bike routes

Citi Bike Live St

NYC BigApps

Data to solve Citi Bike's B

Tags citibike, big apps, city gov

Bike Share Insp

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to service level agreemen

More

	A	B	C	D	E	F	G	H	I	J	K	L	M	I
	ride_id	rideable_type	started_at	ended_at	start_station	start_station	end_station	end_station	start_lat	start_lng	end_lat	end_lng	member_casual	
1	17AE31FCAE	electric_bike	22:55.7	25:09.7	7 St & Monro	HB304	4 St & Grand	HB301	40.7464126	-74.037977	40.742258	-74.035111	member	
2	FD9859BDBI	electric_bike	15:08.6	17:45.0	7 St & Monro	HB304	4 St & Grand	HB301	40.7464126	-74.037977	40.742258	-74.035111	member	
3	AAC5ECD09I	electric_bike	07:27.0	09:38.2	7 St & Monro	HB304	4 St & Grand	HB301	40.7464126	-74.037977	40.742258	-74.035111	member	
4	857C4DCB2	electric_bike	43:18.9	45:38.2	7 St & Monro	HB304	4 St & Grand	HB301	40.7464126	-74.037977	40.742258	-74.035111	member	
5	4439657C24	classic_bike	29:40.2	32:56.9	Clinton St & f	HB409	4 St & Grand	HB301	40.73743	-74.03571	40.742258	-74.035111	member	
6	45EE1276D5	electric_bike	20:57.5	23:24.5	7 St & Monro	HB304	4 St & Grand	HB301	40.7464126	-74.037977	40.742258	-74.035111	member	
7	ED7519417C	electric_bike	16:48.6	19:40.1	7 St & Monro	HB304	4 St & Grand	HB301	40.7464126	-74.037977	40.742258	-74.035111	member	
8	6AE9DF5DDI	electric_bike	25:05.7	27:31.9	7 St & Monro	HB304	4 St & Grand	HB301	40.7464126	-74.037977	40.742258	-74.035111	member	
9	EFA3AF0E8C	classic_bike	33:58.3	35:33.7	Clinton St & f	HB409	4 St & Grand	HB301	40.73743	-74.03571	40.742258	-74.035111	member	
10	1DD6CF59B	classic_bike	20:56.6	24:28.6	7 St & Monro	HB304	4 St & Grand	HB301	40.7464126	-74.037977	40.742258	-74.035111	member	
11	4AFDFFF03E	electric_bike	16:18.6	18:36.5	7 St & Monro	HB304	4 St & Grand	HB301	40.7464126	-74.037977	40.742258	-74.035111	member	
12	47256D9756	electric_bike	14:07.3	24:32.5	Grand St	JC102	Monmouth ai	JC075	40.7151777	-74.037683	40.7256855	-74.04879	member	
13	D6D48EE8F4	classic_bike	24:27.5	32:36.7	Willow Ave &	HB505	4 St & Grand	HB301	40.7518675	-74.030377	40.742258	-74.035111	member	
14	8A4C635C52	classic_bike	01:08.8	07:47.7	Clinton St & f	HB409	4 St & Grand	HB301	40.73743	-74.03571	40.742258	-74.035111	casual	
15	D3B413B66E	electric_bike	07:50.6	09:48.9	Clinton St & f	HB409	4 St & Grand	HB301	40.73743	-74.03571	40.742258	-74.035111	member	
16	D22B09BB07	electric_bike	51:04.1	53:10.9	Clinton St & f	HB409	4 St & Grand	HB301	40.73743	-74.03571	40.742258	-74.035111	member	
17	169A0222C1	classic_bike	00:48.2	04:56.0	Clinton St & f	HB409	4 St & Grand	HB301	40.73743	-74.03571	40.742258	-74.035111	member	
18	D890D46C7I	electric_bike	49:34.9	58:12.2	Baldwin at M	JC020	Monmouth ai	JC075	40.7236589	-74.064194	40.7256855	-74.04879	member	
19	1C8BB3F1Bf	electric_bike	13:49.4	19:49.5	Grand St	JC102	Monmouth ai	JC075	40.7151777	-74.037683	40.7256855	-74.04879	member	
20	A566A5852F	electric_bike	11:41.5	36:15.1	Grand St	JC102	4 St & Grand	HB301	40.7151777	-74.037683	40.742258	-74.035111	member	
21	71E4F5A9BE	classic_bike	24:01.8	48:59.3	Clinton St & f	HB409	Clinton St & f	HB409	40.73743	-74.03571	40.73743	-74.03571	casual	
22	97F3829FB0	classic_bike	30:25.3	57:41.8	Clinton St & f	HB409	Clinton St & f	HB409	40.73743	-74.03571	40.73743	-74.03571	member	
23	AB60573017	classic_bike	11:08.8	28:04.4	Clinton St & f	HB409	Clinton St & f	HB409	40.73743	-74.03571	40.73743	-74.03571	member	
24	D037175CF1	electric_bike	57:18.8	59:24.1	7 St & Monro	HB304	4 St & Grand	HB301	40.7464126	-74.037977	40.742258	-74.035111	member	
25	8F72538505	electric_bike	26:47.1	29:18.3	7 St & Monro	HB304	4 St & Grand	HB301	40.7464126	-74.037977	40.742258	-74.035111	member	
26	F9D102C02C	electric_bike	28:48.9	31:04.0	7 St & Monro	HB304	4 St & Grand	HB301	40.7464126	-74.037977	40.742258	-74.035111	member	
27	62F91D30E6	electric_bike	58:13.8	00:44.2	7 St & Monro	HB304	4 St & Grand	HB301	40.7464126	-74.037977	40.742258	-74.035111	member	
28	5BD562082C	electric_bike	14:34.7	17:00.0	7 St & Monro	HB304	4 St & Grand	HB301	40.7464126	-74.037977	40.742258	-74.035111	member	
29														

Updated
September 10, 2018

Data Provided by
CitiBike

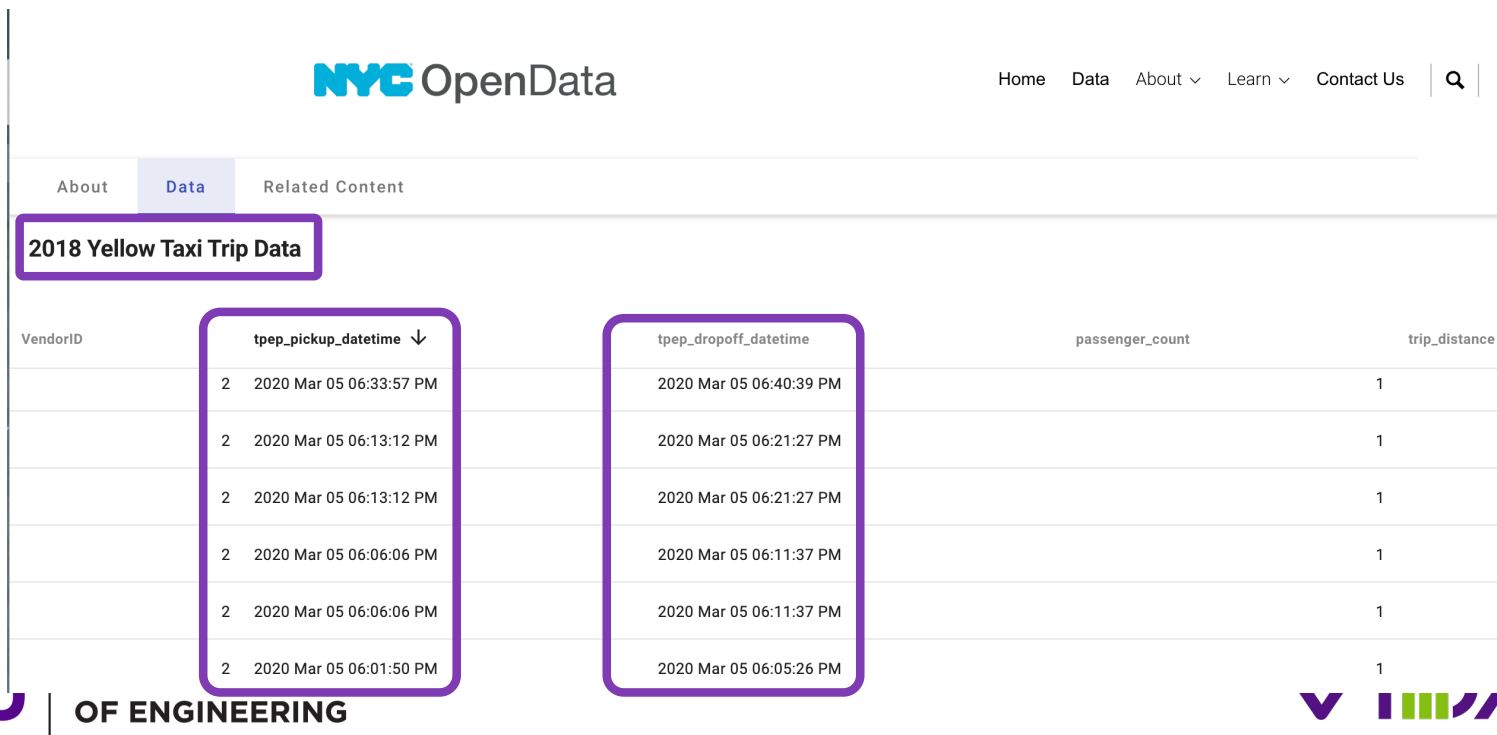
VIA

VISUALIZATION
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Views
1,844

Querying Dataset Collections

- Difficult to express data discovery information needs using keyword-based queries and to assess dataset relevance based on textual snippets
- Metadata is necessarily incomplete and sometimes inconsistent with the data



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About Data Related Content

2018 Yellow Taxi Trip Data

VendorID	tpep_pickup_datetime ↓	tpep_dropoff_datetime	passenger_count	trip_distance
2	2020 Mar 05 06:33:57 PM	2020 Mar 05 06:40:39 PM		1
2	2020 Mar 05 06:13:12 PM	2020 Mar 05 06:21:27 PM		1
2	2020 Mar 05 06:13:12 PM	2020 Mar 05 06:21:27 PM		1
2	2020 Mar 05 06:06:06 PM	2020 Mar 05 06:11:37 PM		1
2	2020 Mar 05 06:06:06 PM	2020 Mar 05 06:11:37 PM		1
2	2020 Mar 05 06:01:50 PM	2020 Mar 05 06:05:26 PM		1



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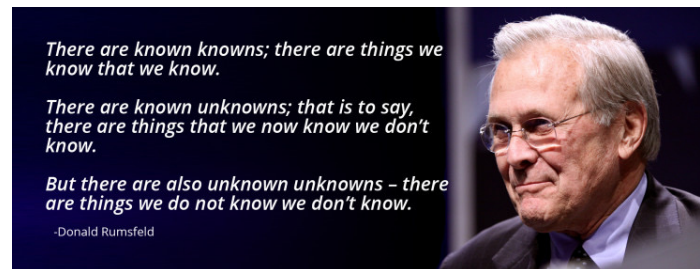
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Querying Dataset Collections

- Difficult to express data discovery information needs using keyword-based queries and to assess dataset relevance based on textual snippets
- Metadata is necessarily incomplete and sometimes inconsistent with the data
- *Mismatch between users' requirements and metadata + search capabilities* [Papenmeier et al., 2021]

“complex information needs seem to collide with the capabilities of data search systems.”

- Search for what you know – limited support for discovering *unknown unknowns*
 - *Similar to the early days of the Web, remember DMOZ?*



Dataset Discovery: Desiderata

- Go beyond the metadata provided by data publishers: leverage dataset **contents and relationships**
- **Informative snippets and result presentation**: facilitate the identification of relevant data
- Better mechanisms to search dataset collections: **expressive queries** that capture diverse information needs

The Auctus Dataset Search Engine

- *Discover data published* on the Web and in a variety of repositories
 - DSDD: emulates a user interacting with a search engine [Zhang et al., ACM CIKM 2021]
- Ingest datasets from multiple repositories, e.g., zenodo, open data for cities and governments
- Go beyond the metadata provided by data publishers -- *leverage dataset contents to enable findability*
- *Index metadata*, allowing users and systems to search for data through a rich query interface that supports diverse information needs
- Explore *new visual representations* and *interaction modes*

<https://auctus.vida-nyu.org/>

[Castelo et al., PVLDB 2021]

<https://www.youtube.com/watch?v=IZQbh3ctq6Q>

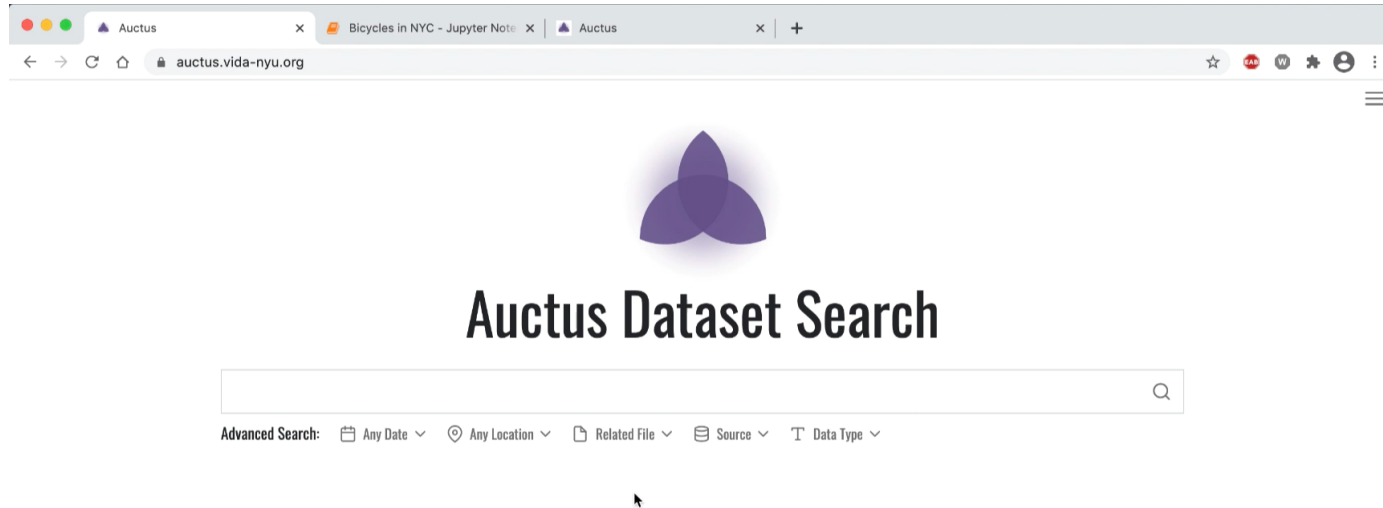


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Auctus in Action



<https://www.youtube.com/watch?v=IZQbh3ctq6Q>



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Inadequate Dataset Descriptions

- Dataset descriptions vary widely
 - 3121 of 23520 datasets (13.2%) on Auctus don't have descriptions
 - 2346 datasets (~10%) have descriptions containing less than 10 words

Citi Bike System Data

NYC BigApps

Data to solve Citi Bike's Big Idea.

Updated
September 10, 2018

Data Provided by
CitiBike

Inadequate Dataset Descriptions

- Dataset descriptions vary widely
 - 3121 of 23520 datasets (13.2%) on Auctus don't have descriptions
 - 2346 datasets (~10%) have descriptions containing less than 10 words
- Google Dataset Search does not index datasets without descriptions
- Negatively impact findability and data relevance assessment
- Users want to know:

- What are the value types and value ranges for attributes?
- What is the timespan of the data?
- What is the geographic region covered by the records?
- What is the spatial and temporal granularity?
- What analyses does the dataset support?
- What is the data about? What is the category/topic?

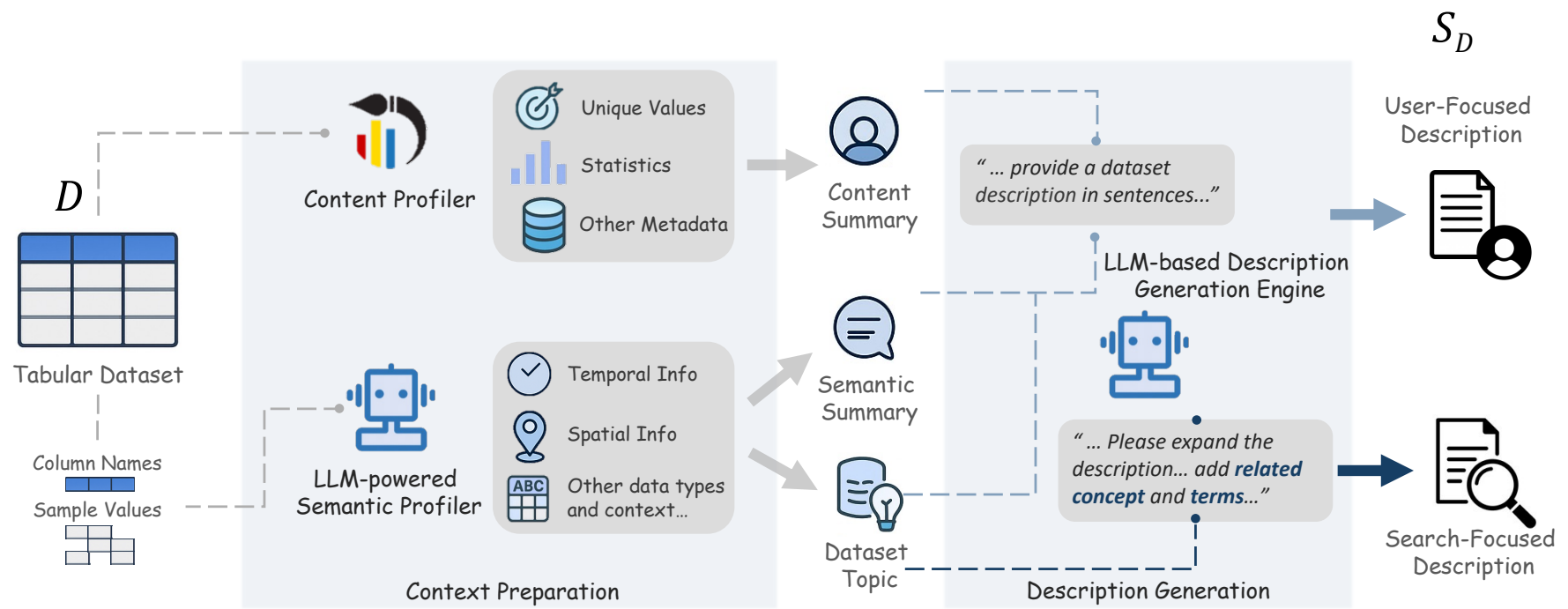
[Koesten et al., JHCS 2020]

LLMs & Data Understanding: Generating Descriptions

- Leverage LLMs' broad world knowledge and their ability of LLMs to produce fluent, readable text
- Generate descriptions that are:
 - Aligned with users' information needs
 - Faithful to the data – accurately describe the dataset contents
 - Concise and readable – facilitate relevance assessment
- Challenges:
 - How to best represent tables in prompts?
 - Reliance on data samples risks deriving incomplete or misleading descriptions that fail to capture crucial global properties, e.g., temporal and spatial coverage
 - Hallucinations

Automatic Dataset Description Generation

$$S_D = \mathcal{G}(D)$$



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[Zhang et al., arxiv 2025]
<https://github.com/VIDA-NYU/AutoDDG>

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Automatic Dataset Description Generation: Benefits

- Improve findability for existing dataset search systems
- Combine LLMs and traditional profiling
 - Use **algorithmic** profilers to summarize the *whole data* – no need to sample
 - Use LLMs to obtain information about dataset semantics
- Small summaries can be efficiently processed by LLMs—reduced cost
- Cost-effective commercial as well as smaller and open LLMs are effective
 - GPT-4o-mini and LLaMA-3.1-8B-Instruct
- Data-driven process reduces the risk for hallucinations
- Trade-off: Conciseness and comprehensiveness
 - *User- vs. search-focused descriptions*

User-Focused Description (UFD)

Prompt for User-Focused Description (UFD)

You are an assistant for a dataset search engine. Your goal is to improve the readability of dataset description for dataset search engine users.

Answer the question using the **dataset sample** below:
D_sample.

Additionally, the **dataset profile** is as follows
D_profile.

Furthermore, the **semantic information** for the dataset columns are *D_semantic*.

Question: Describe the dataset in sentences. Answer:

Example of User-Focused Description (UFD)

This dataset contains wind speed and direction measurements from a specific time period in 2003. The data includes 4433 unique time stamps, with a temporal coverage of May 13 to June 12, 2003, at a resolution of minutes. The average wind speed ranges from 0 to 27.0, with a standard deviation ranging from 0 to 2.78. The average wind direction ranges from 0 to 338.0, with 16 unique values. The dataset provides a comprehensive view of wind patterns during this time period, making it suitable for environmental studies and research.



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Search-Focused Description (SFD)

Prompt for Search-Focused Description (SFD)

You are an assistant for a dataset search engine. Your goal is to improve the performance of the dataset search engine for keyword queries.

You are given a dataset about the **topic** T , with the following **initial description**: *Initial_Description*.

Please expand the description by including the exact topic. Additionally, add as many related concepts, synonyms, and relevant terms as possible based on the initial description and the topic.

Unlike the initial description, which is focused on presentation and readability, the expanded description is intended to be indexed at backend of a dataset search engine to improve searchability.

Therefore, focus less on readability and more on including all relevant terms related to the topic. Make sure to include any variations of the key terms and concepts that could help improve retrieval in search results.

Please follow the structure of following **example template**: *template*

Example of Search-Focused Description (SFD)

Dataset Overview:

- An overview description based on UFD.

Keywords and Themes:

- Wind speed data
- Average wind direction
- Wind patterns over time
- ...

Synonyms and Related Terms:

- Weather data
- Atmospheric data
- Meteorological data
- Climatic data
- ...

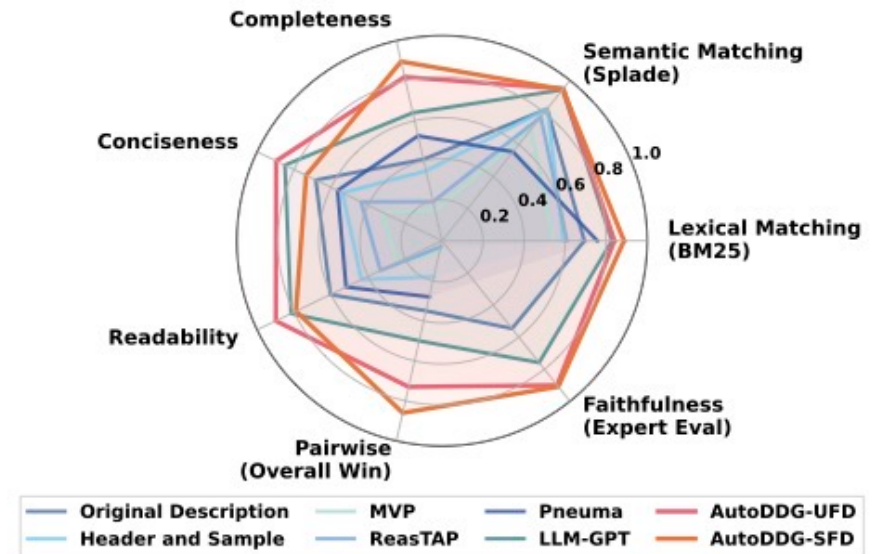
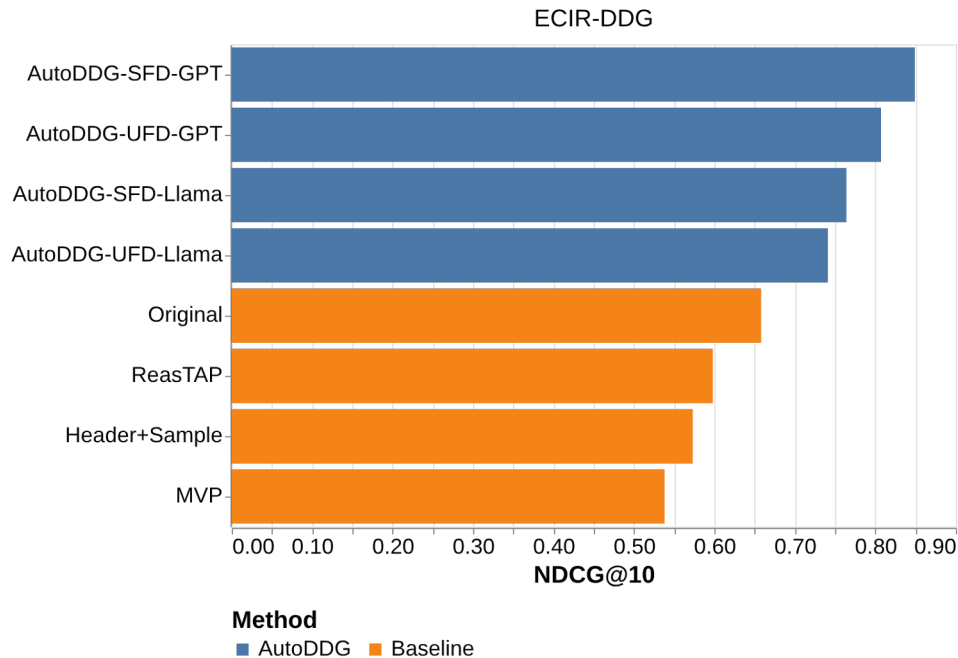
Applications and Use Cases:

- Aids in weather forecasting and atmospheric simulation models.
- Facilitates the development of wind energy projects and assessments of renewable energy potential.
- ...

Automatic Dataset Description Generation: Benefits

- Improve findability for **existing** dataset search systems—just index the descriptions
- Summarizes the whole data – no need to sample
- Small summaries can be efficiently processed by LLMs
- Data-driven process reduces the risk for hallucinations
- Trade-off: Conciseness and comprehensiveness
 - *User- vs. search-focused descriptions*
- Descriptions can be customized for different applications/domains
- Use cases:
 - Authors generate descriptions of their data
 - Repository admin generates descriptions for the dataset collection
- **Dataset description can be understood by LLMs/agents**

Experimental Evaluation



Improves dataset retrieval

Generates high-quality descriptions – compared to to gold standard, judged by an LLM and experts



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Details and ablations in [Zhang et al., arxiv 2025]



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Expert Evaluation: Highlights

- UFDs score higher in conciseness, readability, and semantic coherence while SFDs perform better in completeness—high agreement with LLM judge
- Original descriptions provide only metadata and external links rather than content descriptions, achieving "conciseness at the cost of completeness"
- AutoDDG descriptions are faithful to the data
- LLM-only descriptions frequently contain errors including incorrect temporal ranges and statistics
- The integration of data profilers and topic generators are effective guardrails -- avoiding hallucinations and factual errors

Big Urban Data: What is the **Big** deal?

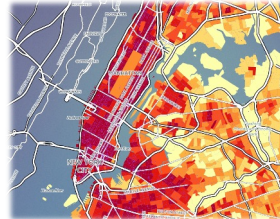
- Cities are the loci of economic activity
- 50% of the world population lives in cities, by 2050 the number will grow to 70%
- Growth leads to problems, e.g., transportation, environment and pollution, housing, infrastructure
- Good news: Lots of data are being collected from traditional and *unsuspecting* sensors

Data Exhaust from Cities

Infrastructure



Environment



People

flickr

twitter



Opportunity: Use data to make cities more efficient and sustainable, and improve the lives of their residents

Big Urban Data: What is hard?

Infrastructure



Condition, operations

Environment



Meteorology, pollution,
noise, flora, fauna

People



Relationships,
economic activities,
health, nutrition, opinions,
...

- City components interact in complex ways over space and time
- Need to analyze the city *data exhaust* to understand these interactions

twitter



DATA.GOV
EMPOWERING PEOPLE

You Tube
Broadcast Yourself

NYC OpenData



data.gov in

Open Government Data Platform India



सत्यमेव जयते

Montréal

PORTAIL DONNÉES
OUVERTES



data.gouv.fr



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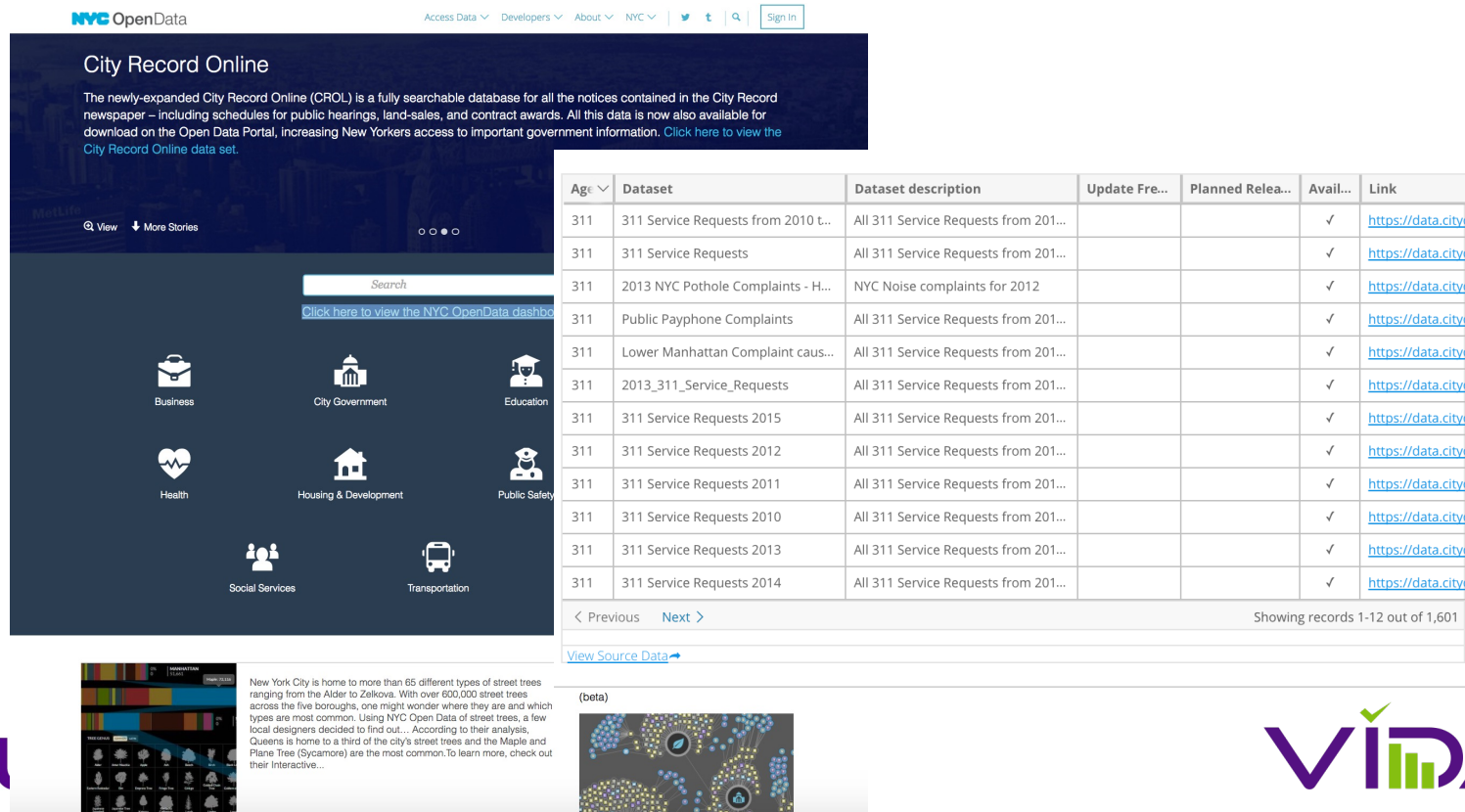


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Big Data Challenge: Data Discovery

- Lots of data sets are available. How to find data relevant to an information need?

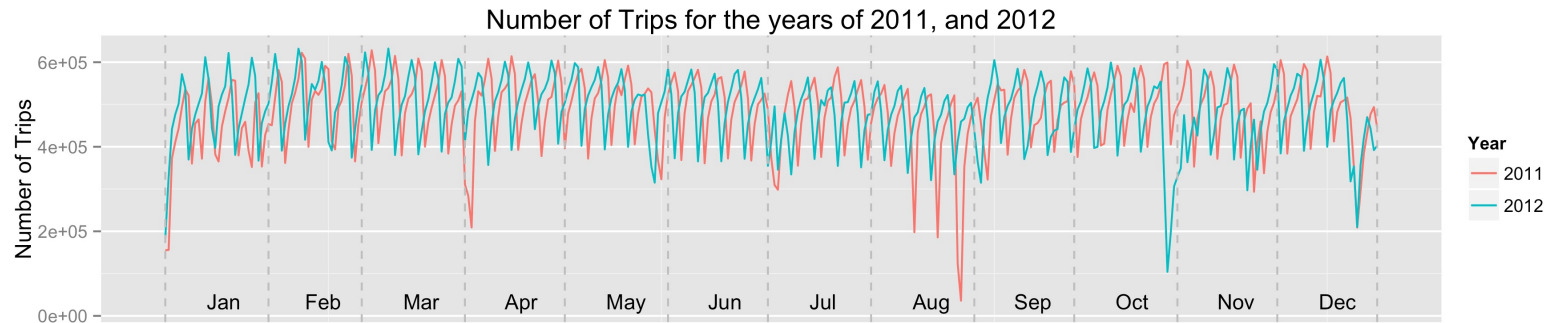


The screenshot displays the NYC OpenData website interface. At the top, the 'NYC OpenData' logo is visible alongside navigation links for 'Access Data', 'Developers', 'About', and 'NYC'. A 'Sign In' button is also present. The main heading is 'City Record Online', followed by a descriptive paragraph about the City Record Online (CROL) database. Below this, there's a search bar and a grid of category icons including Business, City Government, Education, Health, Housing & Development, Public Safety, Social Services, and Transportation. To the right of the website preview, a table lists various datasets available on the platform.

Age	Dataset	Dataset description	Update Fre...	Planned Relea...	Avail...	Link
311	311 Service Requests from 2010 t...	All 311 Service Requests from 201...			✓	https://data.city...
311	311 Service Requests	All 311 Service Requests from 201...			✓	https://data.city...
311	2013 NYC Pothole Complaints - H...	NYC Noise complaints for 2012			✓	https://data.city...
311	Public Payphone Complaints	All 311 Service Requests from 201...			✓	https://data.city...
311	Lower Manhattan Complaint caus...	All 311 Service Requests from 201...			✓	https://data.city...
311	2013_311_Service_Requests	All 311 Service Requests from 201...			✓	https://data.city...
311	311 Service Requests 2015	All 311 Service Requests from 201...			✓	https://data.city...
311	311 Service Requests 2012	All 311 Service Requests from 201...			✓	https://data.city...
311	311 Service Requests 2011	All 311 Service Requests from 201...			✓	https://data.city...
311	311 Service Requests 2010	All 311 Service Requests from 201...			✓	https://data.city...
311	311 Service Requests 2013	All 311 Service Requests from 201...			✓	https://data.city...
311	311 Service Requests 2014	All 311 Service Requests from 201...			✓	https://data.city...

Below the table, there are navigation links for '< Previous' and 'Next >', and a status message 'Showing records 1-12 out of 1,601'. A 'View Source Data' link is also present. At the bottom left, there's a small preview of a data visualization titled 'New York City is home to more than 65 different types of street trees...'. At the bottom right, there's a 'VIDA' logo and the text 'VISUALIZATION IMAGING AND DATA ANALYSIS CENTER'.

Exploring Urban Data: NYC Taxis



Taxis are *sensors* that can provide unprecedented insight into city life: economic activity, human behavior, mobility patterns, ...

“How the taxi fleet activity varies during weekdays?”

“What is the average trip time from Midtown to the airports during weekdays?”

“How was activity in Midtown affected during a presidential visit?”

“How did the movement patterns change during Sandy?”

“Where are the popular night spots?”

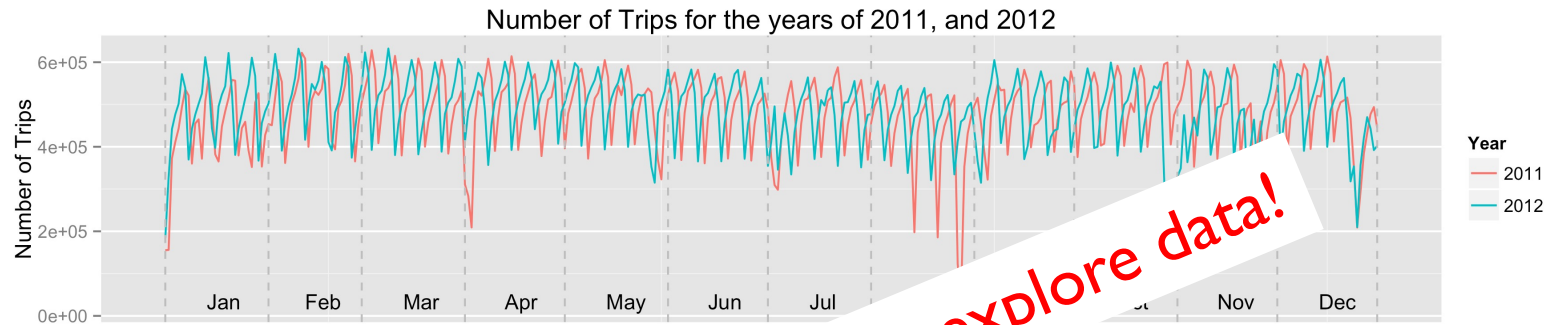


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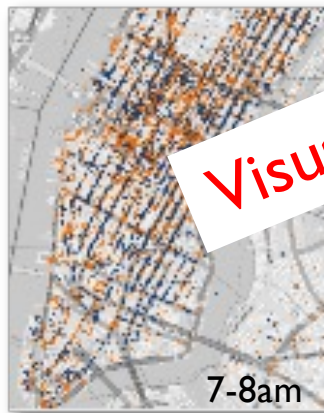
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Exploring Urban Data: NYC Taxis



Visualization is crucial to explore data!

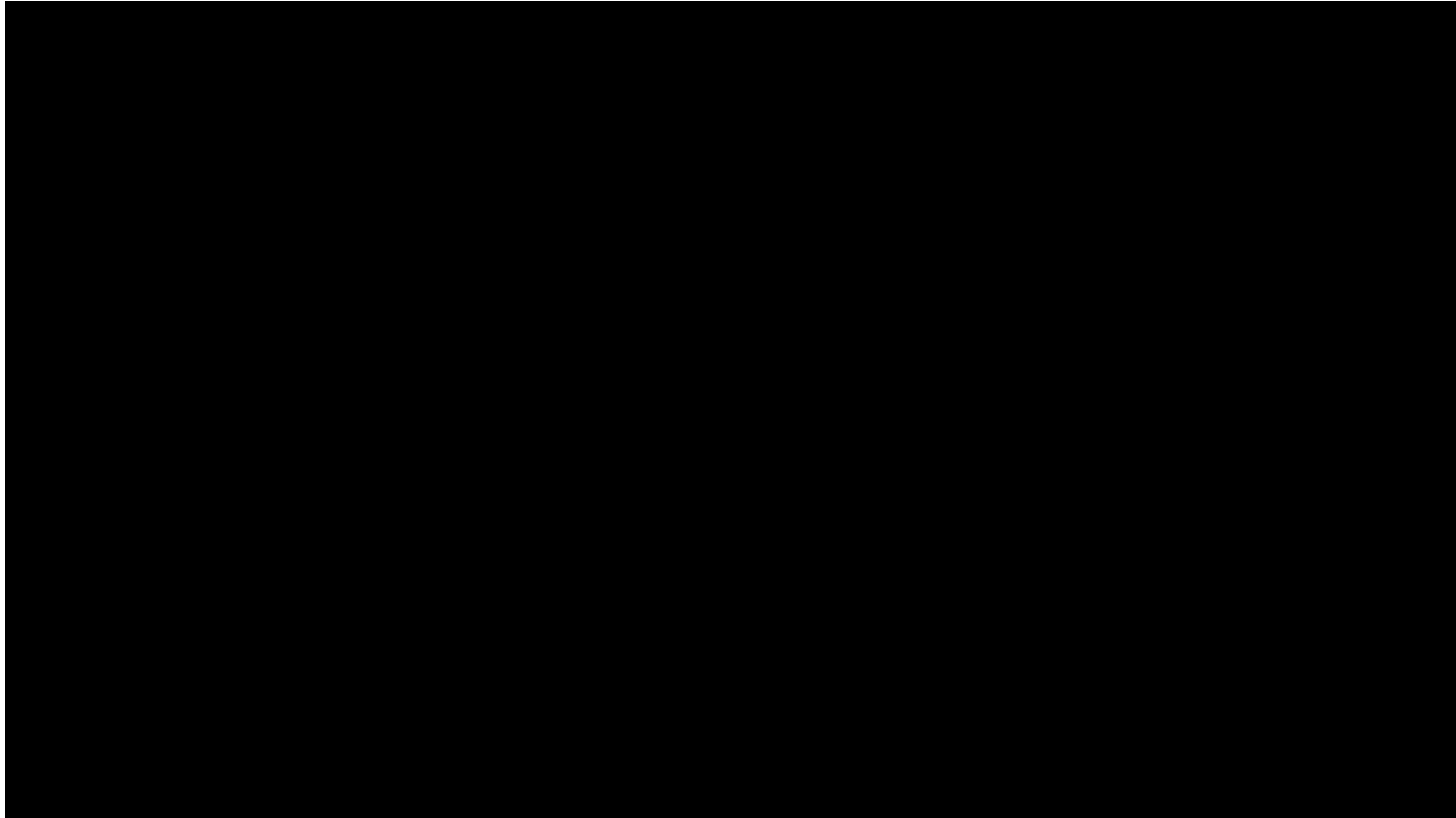


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Exploring Taxi Data: TaxiVis



https://www.youtube.com/watch?v=_B35vxKgDw4&feature=youtu.be



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<https://github.com/ViDA-NYU/TaxiVis>
[Ferreira et al., IEEE TVCG 2013]



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Impact: TaxiVis

----- Forwarded message -----

From: [REDACTED] <[REDACTED]@tlc.nyc.gov>

Date: Thu, Oct 24, 2013 at 4:58 PM

Subject: NYC taxi data

To: "Claudio Silva (csilva@nyu.edu)" <csilva@nyu.edu>, "Huy Vo (huy.vo@nyu.edu)" <huy.vo@nyu.edu>, "Caryn Joy Knutsen (caryn.knutsen@nyu.edu)" <caryn.knutsen@nyu.edu>, "Kim Alfred (kim.alfred@nyu.edu)" <kim.alfred@nyu.edu>



Hi all,

First, ~~I would like to thank you all~~ for coming to TLC data. We were truly blown away! In fact, we had been looking for a product like the one you've demonstrated to us. After

[REDACTED]
for us on Monday. We think that could be a great start for future use for our data in combination with other available

"The speed at which the tool permits us to work has saved multiple hours of staff time and has dramatically improved the unit's output and capabilities."

Assistant Commissioner, DoT

Cheers,

[REDACTED]



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<http://www.taxivis.org>

[Ferreira et al., IEEE TVCG 2013]



Urban Data Quality

- NYC Taxi Data: ~500k trips/day; 868 million trips in 5 years

Dataset	Statistic	Trip Duration (min)	Trip Distance (mi)	Fare Amount (US\$)	Tip Amount (US\$)
2008	Min	0.00	0.00	0.00	0.00
	Avg	16.74	2.71	0.09	0.10
	Max	1440.00	50.00	10.00	8.75
2009	Min	0.00	0.00	2.50	0.00
	Avg	7.75	6.22	6.04	0.38
	Max	180.00	180.00	200.00	200.00
2010	Min	-1,760.00	-21,474,834.00	-21,474,808.00	-1,677,720.10
	Avg	6.76	5.89	9.84	2.11
	Max	1,322.00	16,201,631.40	93,960.07	938.02
2011	Min	0.00	0.00	2.50	0.00
	Avg	12.35	2.80	10.25	2.22
	Max	180.00	100.00	500.00	200.00
2012	Min	0.00	0.00	2.50	0.00
	Avg	12.32	2.88	10.96	2.32
	Max	180.00	100.00	500.00	200.00

NYC Taxi Data quality issues [Freire et al., IEEE DEB
2016]



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Challenge: Clean and Curate

Trips with 255 passengers
Very large tips, >\$1000
Trips missing pickup or dropoff location
No tips for trips paid by cash
Trips starting or ending in the river

Bad data → bad analysis results



NYC Taxi Data quality issues [Freire et al., IEEE DEB
2016]

Challenge: *Interactive* Query Evaluation

- Requirement: support interactive queries
- Raw taxi data:
 - 3 years
 - 150 GB in 48 CSV files
 - 520M trips
 - 12 fields, 2 spatial-temporal attributes
- After ETL: 50 GB in binary format

	SQLite	Postgre SQL
Storage Space in GB	100	200
Building Indices in Minutes (One Year of Data)	3,120	780
1K Items Query in	8	3
Seconds		24

*“increased latency **reduces** the rate at which users make observations, draw generalizations and generate hypotheses”*



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[Liu and Heer, IEEE TVCG 2014]



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Supporting Interactive Queries

Solution 1: New in-memory spatio-temporal index based on kd-trees

- Can index multiple attributes!
- Tree nodes store *kd-tree*
- A leaf node represents a k-dimensional node that satisfies the path constraints

	SQLite	PostgreSQL	Our Solution
Storage Space in GB	100	200	30
Building Indices in Minutes (One Year of Data)	3,120	780	28
1K Items Query in Seconds	8	3	0.2
100K Items Query in Seconds	85	24	2

[Ferreira et al., IEEE TVCG 2013]



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Supporting Interactive Queries

Solution 2: Spatio-temporal index based on out-of-core kd-tree using GPUs

- Can index multiple attributes!
- Tree nodes store k-d tree
- Leaf nodes represent a *set of k-dimensional nodes*
 - Point to a leaf block containing records that satisfy the path constraints
 - Store the bounding box for the records

Query	MongoDB (1 GPU)	MongoDB (3 GPUs)	PostgreSQL			ComDB		
	Time(sec)	Time(sec)	Time(sec)	Speedup (1 GPU)	Speedup (3 GPUs)	Time(sec)	Speedup (1 GPU)	Speedup (3 GPUs)
1	0.237	0.103	141.8	598	1376	136.9	578	1329
2	0.199	0.065	129.2	649	1987	119.6	601	1840
3	0.202	0.093	97.1	480	1044	39.4	195	423
4	0.183	0.069	103.7	566	1502	25.6	140	371
5	0.361	0.159	106.3	294	668	23.8	66	149
6	0.325	0.174	102.6	315	589	28.9	89	166

2 orders of magnitude faster than RDBMSs



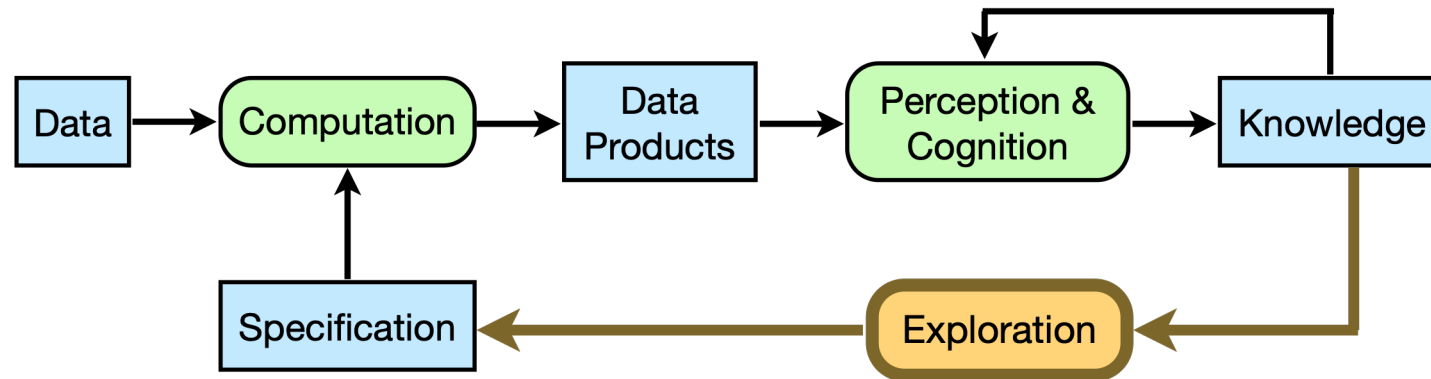
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[Doraiswamy et al., IEEE ICDE 2016]



Challenge: Transparency, Reproducibility, Trust

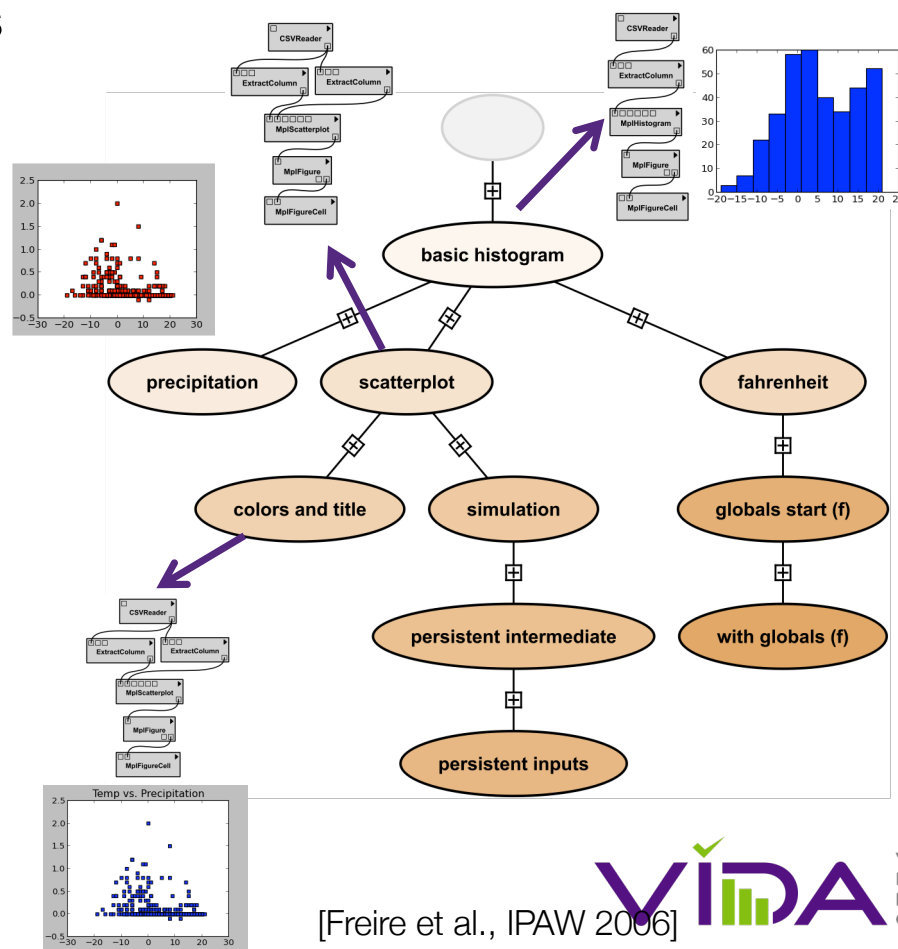


[Modified from Van Wijk, Vis 2005]

- Iterative process to generate and test hypotheses
- Easy to get lost – derive a result and not remember how you got there
 - Lots of data: both input and derived
 - Complex processes that encompass multiple tools
- Need detailed provenance of the exploratory process
 - Provenance enables transparency, reproducibility, and trust

Provenance of Data Exploration

- Keep track of code and data versions
- Debug!



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[Freire et al., IPAW 2006]



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Debugging Data and Computations



<https://arstechnica.com/information-technology/2019/10/chemists-discover-cross-platform-python-scripts-not-so-cross-platform/>

OUT OF SORTS —

Researchers find bug in Python script may have affected hundreds of studies

“The scripts [...] were found to return correct results on macOS Mavericks and Windows 10. But on macOS Mojave and Ubuntu, the results were off by nearly a full percent.”

- Scripts used a specific library, *glob*, which returns a different sorted order depending on the OS
- Problem is not easy to detect



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Big Data: What is hard? What will we learn?

- There is no one-size-fits-all solution + lots of tools + rapidly-evolving technology:
often, *you have to build your own*
Need to understand **foundations**
- Scalability can be challenging
Learn techniques from distributed systems, parallel databases, big data stack – do not reinvent the wheel!
- Leverage new architectures: clouds, GPUs, multiple cores
Different computation models: need new algorithms
- Lots of diverse and dirty data
Use and combine different cleaning and integration techniques (inc. AI-powered methods)
- Transparency and trust
Apply transparency and reproducibility best practices, debug data/computations

Merci
Thank you
Obrigada
благодаря
Kiitos
धन्यवाद
Tack
Danke
Ευχαριστω
Bedankt



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